Constant-dimensional generalized Sylvester-Gallai configurations of quadratic polynomials

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Allowed standard prerequisites (editorial).

- Robust Sylvester-Gallai theorem and standard algebraic tools. Original CCC Section2 definitions/statements retained, including robust theorem13[9], resultant17[8], Nullstellensatz and UFD usage; no recursive proof of external standard results.

- Small product-in-radical witness bound. CCC Claim35 and journal Claim3.2 cite original Gupta Claim11, with author statement retained.

Theorem 4. *There exists a universal constant c such that the following holds. Let $\widetilde{\mathcal{Q}} = \{Q_i\}_{i\in\{1,\ldots,m\}} \subset \mathbb{C}[x_1,\ldots,x_n]$ be a finite set of pairwise linearly independent homogeneous polynomials, such that every $Q_i \in \widetilde{\mathcal{Q}}$ is either irreducible or a square of a linear form. Assume that, for every $i \neq j$, whenever Q_i and Q_j vanish then so does $\prod_{k\in\{1,\ldots,m\}\setminus\{i,j\}} Q_k$. Then,* $\dim(\mathrm{span}\{\mathcal{Q}\}) \leq c$.

Theorem 5. *Let $\{Q_k\}_{k\in\mathcal{K}}, A, B$ be homogeneous polynomials of degree2 such that $\prod_{k\in\mathcal{K}} Q_k \in \sqrt{\langle A,B\rangle}$. Then one of the following cases hold:*

(i) There is $k \in \mathcal{K}$ such that Q_k is in the linear span of A, B.

(ii) There exists a non trivial linear combination of the form $\alpha A + \beta B = c \cdot d$ where c and d are linear forms.

(iii) There exist two linear forms c and d such that when setting $c = d = 0$ we get that A, B and one of $\{Q_k\}_{k\in\mathcal{K}}$ vanish.

From now on, to ease notations, we use Theorem5i, Theorem5ii or Theorem5iii to describe different cases of Theorem5. The CCC formal version p8:5 refers the complete proof to its full version. The complete published author proof of precisely this structural statement is included below from journal Theorem3.1, pp11–16, CC BY4.0; the CCC main proof retains its own parameters and notation.

2 Preliminaries

In this section we explain our notation and present some basic algebraic preliminaries.

We will use the following notation. Greek letters $\alpha, \beta, \ldots$ denote scalars from $\mathbb{C}$. Non-capitalized letters $a, b, c, \ldots$ denote linear forms and x, y, z denote variables (which are also linear forms). Bold faced letters denote vectors, e.g. $\mathbf{x} = (x_1, \ldots, x_n)$ denotes a vector of variables, $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_n)$ is a vector of scalars, and $\mathbf{0} = (0, \ldots, 0)$ the zero vector. We sometimes do not use a boldface notation for a point in a vector space if we do not use its structure as vector. Capital letters such as A, Q, P denote quadratic polynomials whereas V, U, W denote linear spaces. Calligraphic letters $\mathcal{I}, \mathcal{J}, \mathcal{F}, \mathcal{Q}, \mathcal{T}$ denote sets. For a positive integer n we denote $[n] = \{1, 2, \ldots, n\}$. For a matrix X we denote by $|X|$ the determinant of X.

A *Commutative Ring* is a group that is abelian with respect to both multiplication and addition operations. We mainly use the multivariate polynomial ring, $\mathbb{C}[x_1, \ldots, x_n]$. An *Ideal* $I \subseteq \mathbb{C}[x_1, \ldots, x_n]$ is an abelian subgroup that is closed under multiplication by ring elements. For $\mathcal{S} \subset \mathbb{C}[x_1, \ldots, x_n]$, we denote with $\langle\mathcal{S}\rangle$, the ideal generated by $\mathcal{S}$, that is, the smallest ideal that contains $\mathcal{S}$. For example, for two polynomials Q_1 and Q_2, the ideal $\langle Q_1, Q_2\rangle$ is the set $\mathbb{C}[x_1, \ldots, x_n]Q_1 + \mathbb{C}[x_1, \ldots, x_n]Q_2$. For a linear subspace V, we have that $\langle V\rangle$ is the ideal generated by any basis of V. The *radical* of an ideal I, denoted

by $\sqrt{I}$, is the set of all ring elements, r, satisfying that for some natural number m (that may depend on r), $r^m \in I$. Hilbert's Nullstellensatz implies that, in $\mathbb{C}[x_1, \ldots, x_n]$, if a polynomial Q vanishes whenever Q_1 and Q_2 vanish, then $Q \in \sqrt{\langle Q_1, Q_2 \rangle}$ (see e.g. [8]). We shall often use the notation $Q \in \sqrt{\langle Q_1, Q_2 \rangle}$ to denote this vanishing condition. For an ideal $I \subseteq \mathbb{C}[x_1, \ldots, x_n]$ we denote by $\mathbb{C}[x_1, \ldots, x_n]/I$ the *quotient ring*, that is, the ring whose elements are the cosets of I in $\mathbb{C}[x_1, \ldots, x_n]$ with the proper multiplication and addition operations. For an ideal $I \subseteq \mathbb{C}[x_1, \ldots, x_n]$ we denote the set of all common zeros of elements of I by $Z(I)$.

For $V_1, \ldots, V_k$ linear spaces, we use $\sum_{i=1}^{k} V_i$ to denote the linear space $V_1 + \ldots + V_k$. For two non zero polynomials A and B we denote $A \sim B$ if $B \in \text{span}\{A\}$. For a space of linear forms $V = \text{span}\{v_1, \ldots, v_\Delta\}$, we say that a polynomial $P \in \mathbb{C}[x_1, \ldots, x_n]$ depends only on V if the value of P is determined by the values of the linear forms $v_1, \ldots, v_\Delta$. More formally, we say that P depends only on V if there is a Δ-variate polynomial $\tilde{P}$ such that $P \equiv \tilde{P}(v_1, \ldots, v_\Delta)$. We denote by $\mathbb{C}[v_1, \ldots, v_\Delta] \subseteq \mathbb{C}[x_1, \ldots, x_n]$ the subring of polynomials that depend only on V.

Another notation that we will use throughout the proof is congruence modulo linear forms.

Definition 7. *Let $V \subset \mathbb{C}[x_1, \ldots, x_n]$ be a space of linear forms, and $P, Q \in \mathbb{C}[x_1, \ldots, x_n]$. We say that $P \equiv_V Q$ if $P - Q \in \langle V \rangle$.*

Fact 8. *Let $V \subset \mathbb{C}[x_1, \ldots, x_n]$ be a space of linear forms and $P, Q \in \mathbb{C}[x_1, \ldots, x_n]$. If $P = \prod_{k=1}^{t} P_k$, and $Q = \prod_{k=1}^{t} Q_k$ satisfy that for all k, P_k and Q_k are irreducible in $\mathbb{C}[x_1, \ldots, x_n]/\langle V \rangle$, and $P \equiv_V Q \not\equiv_V 0$ then, up to a permutation of the indices, $P_k \equiv_V Q_k$ for all $k \in [t]$.*

This follows from the fact that the quotient ring $\mathbb{C}[x_1, \ldots, x_n]/\langle V \rangle$is a unique factorization domain.

2.1 Sylvester-Gallai Theorem and some of its Variants

In this section we present the formal statement the of Sylvester-Gallai theorem and the extensions that we use in this work.

Definition 9. *Given a set of points, $v_1, \ldots, v_m$, we call a line that passes through exactly two of the points of the set an* ordinary line.

Theorem 10 (Sylvester-Gallai theorem)**.** *If m distinct points $v_1, \ldots, v_m$ in $\mathbb{R}^n$ are not collinear, then they define at least one ordinary line.*

Theorem 11 (Kelly's theorem)**.** *If m distinct points $v_1, \ldots, v_m$ in $\mathbb{C}^n$ are not coplanar, then they define at least one ordinary line.*

The robust version of the theorem was stated and proved in [3, 9].

Definition 12. *We say that a set of points $v_1, \ldots, v_m \in \mathbb{C}^n$ is a δ-SG configuration if for every $i \in [m]$ there exists at least δm values of $j \in [m]$ such that the line through v_i, v_j contains a third point in the set.*

Theorem 13 (Robust Sylvester-Gallai theorem, Theorem 1.9 of [9])**.** *Let $V = \{v_1, \ldots, v_m\} \subset \mathbb{C}^n$ be a δ-SG configuration. Then* $\dim(\text{span}\{v_1, \ldots, v_m\}) \leq \frac{12}{\delta} + 1$.

The following is the colored version of the Sylvester-Gallai theorem.

Theorem 14 (Theorem 3 of [12]). *Let $\mathcal{T}_i$, for $i \in [3]$, be disjoint finite subsets of $\mathbb{C}^n$ such that for every $i \neq j$ and any two points $p_1 \in \mathcal{T}_i$ and $p_2 \in \mathcal{T}_j$ there exists a point p_3 in the third set that lies on the line passing through p_1 and p_2. Then, any such $\mathcal{T}_i$ satisfy that* $\dim(\text{span}\{\cup_i \mathcal{T}_i\}) \leq 3$.

We also state the equivalent algebraic versions of Sylvester-Gallai.

Theorem 15. *Let $\mathcal{S} = \{\mathbf{s}_1, \ldots, \mathbf{s}_m\} \subset \mathbb{C}^n$ be a set of pairwise linearly independent vectors such that for every $i \neq j \in [m]$ there is a distinct $k \in [m]$ for which $\mathbf{s}_k \in \text{span}\{\mathbf{s}_i, \mathbf{s}_j\}$. Then* $\dim(\mathcal{S}) \leq 3$.

Theorem 16. *Let $\mathcal{P} = \{\ell_1, \ldots, \ell_m\} \subset \mathbb{C}[x_1, \ldots, x_n]$ be a set of pairwise linearly independent linear forms such that for every $i \neq j \in [m]$ there is a distinct $k \in [m]$ for which whenever ℓ_i, ℓ_j vanish so does ℓ_k. Then* $\dim(\mathcal{P}) \leq 3$.

In this paper we refer to each of Theorem 11, Theorem 15 and Theorem 16 as the Sylvester-Gallai theorem. We shall also refer to sets of points/vectors/linear forms that satisfy the conditions of the relevant theorem as satisfying the condition of the Sylvester-Gallai theorem.

2.2 Resultant

A tool that will play an important role in the proof of Theorem 5 is the resultant of two polynomials. We will only define the resultant of a a quadratic polynomial and a linear polynomial as this is the case relevant to our work.[1] Let $A, B \in \mathbb{C}[x_1, \ldots, x_n]$. View A and B as polynomials in x_1 over $\mathbb{C}[x_2, \ldots, x_n]$ and assume that $\deg_{x_1}(A) = 2$ and $\deg_{x_1}(B) = 1$, namely,

$$A = \alpha x_1^2 + a x_1 + A_0 \quad \text{and } B = b x_1 + B_0 \ .$$

Then, the resultant of A and B with respect to x_1 is the determinant of their Sylvester matrix

$$\text{Res}_{x_1}(A, B) := \begin{vmatrix} A_0 & B_0 & 0 \\ a & b & B_0 \\ \alpha & 0 & b \end{vmatrix} .$$

A useful fact is that if the resultant of A and B vanishes then they share a common factor.

Theorem 17 (See e.g. Proposition 8 in §5 of Chapter 3 in [8]). *Given $F, G \in \mathbb{F}[x_1, \ldots, x_n]$ of positive degree in x_1, the resultant* $\text{Res}_{x_1}(F, G)$ *is an integer polynomial in the coefficients of F and G. Furthermore, F and G have a common factor in $\mathbb{F}[x_1, \ldots, x_n]$ if and only if* $\text{Res}_{x_1}(F, G) = 0$.

2.3 Rank of Quadratic Polynomials

In this section we define the rank of a quadratic polynomial, and present some of its useful properties.

[1] For the general definition of Resultant, see Definition 2 in §5 of Chapter 3 in [8].

Definition 18. *For a homogeneous quadratic polynomial Q we denote with* $\text{rank}_s(Q)$ *the minimal r such that there are $2r$ linear forms $\{a_k\}_{k=1}^{2r}$ satisfying $Q = \sum_{k=1}^{r} a_{2k} \cdot a_{2k-1}$. We call such representation a* minimal representation *of Q.*

This is a slightly different definition than the usual way one defines rank of quadratic forms,[2] but it is more suitable for our needs. We note that a quadratic Q is irreducible if and only if $\text{rank}_s(Q) > 1$. The next claim shows that a minimal representation is unique in the sense that the space spanned by the linear forms in it is unique.

Claim 19. *Let Q be a homogeneous quadratic polynomial and let $Q = \sum_{i=1}^{r} a_{2i-1} \cdot a_{2i}$ and $Q = \sum_{i=1}^{r} b_{2i-1} \cdot b_{2i}$ be two different minimal representations of Q. Then* $\text{span}\{a_1, \ldots, a_{2r}\} = \text{span}\{b_1, \ldots, b_{2r}\}$.

Proof. Note that if the statement does not hold then, without loss of generality, a_1 is not contained in the span of the b_i's. This means that when setting $a_1 = 0$ the b_i's are not affected on the one hand, thus Q remains the same function of the b_i's, and in particular $\text{rank}_s(Q|_{a_1=0}) = r$, but on the other hand $\text{rank}_s(Q|_{a_1=0}) = r - 1$ (when considering its representation with the a_i's), in contradiction. □

This claim allows us to define the notion of *minimal space* of a quadratic polynomial Q, which we shall denote $\text{Lin}(Q)$.

Definition 20. *Let Q be a quadratic polynomial, where* $\text{rank}_s(Q) = r$*, and let $Q = \sum_{i=1}^{r} a_{2i-1}a_{2i}$ be some minimal representation of Q. Define $Lin(Q) :=$* $\text{span}\{a_1, \ldots, a_{2r}\}$*, also denote $Lin(Q_1, \ldots, Q_k) = \sum_{i=1}^{k} Lin(Q_i)$.*

Theorem 19 shows that the minimal space is well defined. The following fact is easy to verify.

Fact 21. *Let $Q = \sum_{i=1}^{m} a_{2i-1} \cdot a_{2i}$ be a homogeneous quadratic polynomial, then $Lin(Q) \subseteq \text{span}\{a_1, \ldots, a_{2m}\}$.*

We now give some basic claims regarding rank_s.

Claim 22. *Let Q be a homogeneous quadratic polynomial with* $\text{rank}_s(Q) = r$*, and let $V \subset \mathbb{C}[x_1, \ldots, x_n]$ be a linear space of linear forms such that* $\dim(V) = \Delta$*. Then* $\text{rank}_s(Q|_{V=0}) \geq r - \Delta$.

Proof. Assume without loss of generality $V = \text{span}\{x_1, \ldots, x_\Delta\}$, and consider $Q \in \mathbb{C}[x_{\Delta+1}, \ldots, x_n][x_1, \ldots, x_\Delta]$. There are $a_1, \ldots, a_\Delta \in \mathbb{C}[x_1, \ldots, x_n]$ and $Q' \in \mathbb{C}[x_{\Delta+1}, \ldots, x_n]$ such that $Q = \sum_{i=1}^{\Delta} a_i x_i + Q'$, where $Q|_{V=0} = Q'$. As $\text{rank}_s(\sum_{i=1}^{\Delta} a_i x_i) \leq \Delta$, it must be that $\text{rank}_s(Q|_{V=0}) \geq r - \Delta$. □

Claim 23. *Let $P_1 \in \mathbb{C}[x_1, \ldots, x_k]$, and $P_2 = y_1 y_2 \in \mathbb{C}[y_1, \ldots, y_2]$. Then* $\text{rank}_s(P_1 + P_2) = \text{rank}_s(P_1) + 1$*. Moreover, $y_1, y_2 \in Lin(P_1 + P_2)$.*

[2] $\text{rank}(Q)$ is the minimal t such that there are t linear forms $\{a_k\}_{k=1}^{t}$, satisfying $Q = \sum_{k=1}^{t} a_k^2$.

Proof. Denote $\text{rank}_s(P_1) = r$ and assume towards a contradiction that there are $a_1, \ldots, a_{2r}$ linear forms in $\mathbb{C}[x_1, \ldots, x_k, y_1, y_2]$ such that $P_1 + P_2 = \sum_{i=1}^{r} a_{2i-1}a_{2i}$. Clearly, $\sum_{i=1}^{r} a_{2i-1}a_{2i} \equiv_{y_1} P_1$. As $\text{rank}_s(P_1) = r$ this is a minimal representation of P_1. Hence, for every i, $a_i|_{y_1=0} \in \text{Lin}(P_1) \subset \mathbb{C}[x_1, \ldots, x_k]$. Moreover, from the minimality of r, $a_i|_{y_1=0} \neq 0$. Therefore, as y_1 and y_2 are linearly independent, we deduce that all the coefficients of y_2 in all the a_i's are 0. By reversing the roles of y_1 and y_2 we can conclude that $a_1, \ldots, a_{2r} \subset \mathbb{C}[x_1, \ldots, x_k]$ which means that Q does not depend on y_1 and y_2 in contradiction. Consider a minimal representation $P_1 = \sum_{i=1}^{2r} b_{2i-1}b_{2i}$, from the fact that $\text{rank}_s(P_1 + P_2) = r + 1$ it follows that $P_1 + P_2 = \sum_{i=1}^{2r} b_{2i-1}b_{2i} + y_1y_2$ is a minimal representation of $P_1 + P_2$ and thus $\text{Lin}(P_1 + P_2) = \text{Lin}(P_1) + \text{span}\{y_1, y_2\}$. □

Corollary 24. *Let a and b be linearly independent linear forms. Then, if c, d, e and f are linear forms such that $ab+cd = ef$ then* $\dim(\text{span}\{a, b\} \cap \text{span}\{c, d\}) \geq 1$.

Claim 25. *Let a, b, c and d be linear forms, and V be a linear space of linear forms. Assume $\{0\} \neq Lin(ab - cd) \subseteq V$ then* $\text{span}\{a, b\} \cap V \neq \{0\}$.

Proof. As $\text{Lin}(ab - cd) \subseteq V$ it follows that $ab \equiv_V cd$. If both sides are zero then $ab \in \langle V \rangle$ and without loss of generality $b \in V$ and the statement holds. If neither sides is zero then from Theorem 8 there are linear forms $v_1, v_2 \in V$, and $\lambda_1, \lambda_2 \in \mathbb{C}^\times$ such that, $\lambda_1\lambda_2 = 1$ and without loss of generality $c = \lambda_1 a + v_1, d = \lambda_2 b + v_2$. Note that not both v_1, v_2 are zero, as $ab - cd \neq 0$. Thus,

$$ab - cd = ab - (\lambda_1 a + v_1)(\lambda_2 b + v_2) = \lambda_1 a v_2 + \lambda_2 b v_1 + v_1 v_2.$$

As $\text{Lin}(ab - cd) \subseteq V$ it follows that $\text{Lin}(\lambda_1 a v_2 + \lambda_2 b v_1) \subseteq V$ and therefore there is a linear combination of a, b in V and the statement holds. □

We end this section with claims that will be useful in our proofs.

Claim 26. *Let $V = \sum_{i=1}^{m} V_i$ where V_i are linear subspaces, and for every i,* $\dim(V_i) = 2$. *If for every $i \neq j \in [m]$,* $\dim(V_i \cap V_j) = 1$, *then either* $\dim(\bigcap_{i=1}^{m} V_i) = 1$ *or* $\dim(V) = 3$.

Proof. Let $w \in V_1 \cap V_2$. Complete it to basis of V_1 and V_2: $V_1 = \text{span}\{u_1, w\}$ and $V_2 = \text{span}\{u_2, w\}$. Assume that $\dim(\bigcap_{i=1}^{m} V_i) = 0$. Then, there is some i for which $w \notin V_i$. Let $x_1 \in V_i \cap V_1$, and so $x_1 = \alpha_1 u_1 + \beta_1 w$, where $\alpha_1 \neq 0$. Similarly, let $x_2 \in V_i \cap V_2$. Since $w \notin V_i$, $x_2 = \alpha_2 u_2 + \beta_2 w$, where $\alpha_2 \neq 0$. Note that $x_1 \notin \text{span}\{x_2\}$, as $\dim(V_1 \cap V_2) = 1$, and w is already in their intersection. Thus, we have $V_i = \text{span}\{x_1, x_2\} \subset \text{span}\{w, u_1, u_2\}$.

Now, consider any other $j \in [m]$. If V_j does not contain w, we can apply the same argument as we did for V_i and conclude that $V_j \subset \text{span}\{w, u_1, u_2\}$. On the other hand, if $w \in V_j$, then let $x_j \in V_i \cap V_j$, it is easy to see that x_j, w are linearly independent and so $V_j = \text{span}\{w, x_j\} \subset \text{span}\{w, V_i\} \subseteq \text{span}\{w, u_1, u_2\}$. Thus, in any case $V_j \subset \text{span}\{w, u_1, u_2\}$. In particular, $\sum_j V_j \subseteq \text{span}\{w, u_1, u_2\}$ as claimed. □

2.4 Projection Mappings

In this section we present and apply a new technique which allows us to simplify the structure of quadratic polynomials. Naively, when we want to simplify a polynomial equation, we can project it on a subset of the variables. Unfortunately, this projection does not necessarily preserve pairwise linear independence, which is a crucial property in our proofs. To remedy this fact, we present a set of mappings, which are somewhat similar to projections, but do preserve pairwise linear independence among polynomials.

Definition 27. *Let $V = \text{span}\{v_1, \ldots, v_\Delta\} \subseteq \text{span}\{x_1, \ldots, x_n\}$ be a Δ-dimensional linear space of linear forms, and let $\{u_1, \ldots, u_{n-\Delta}\}$ be a basis for $V^\perp$. For $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_\Delta) \in \mathbb{C}^\Delta$ we define $T_{\boldsymbol{\alpha},V} : \mathbb{C}[x_1, \ldots, x_n] \mapsto \mathbb{C}[x_1, \ldots, x_n, z]$, where z is a new variable, to be the linear map given by the following action on the basis vectors: $T_{\boldsymbol{\alpha},V}(v_i) = \alpha_i z$ and $T_{\boldsymbol{\alpha},V}(u_i) = u_i$.*

Observation 28. *$T_{\boldsymbol{\alpha},V}$ is a linear transformation and is also a ring homomorphism. This follows from the fact that a basis for $\text{span}\{x_1, \ldots, x_n\}$ is a basis for $\mathbb{C}[x_1, \ldots, x_n]$ as $\mathbb{C}$-algebra.*

Claim 29. *Let $V \subseteq \text{span}\{x_1, \ldots, x_n\}$ be a Δ-dimensional linear space of linear forms. Let F and G be two polynomials that share no common irreducible factor. Then, with probability 1 over the choice of $\boldsymbol{\alpha} \in [0,1]^\Delta$ (say according to the uniform distribution), $T_{\boldsymbol{\alpha},V}(F)$ and $T_{\boldsymbol{\alpha},V}(G)$ do not share a common factor that is not a polynomial in z.*

Proof. Let $\{u_1, \ldots, u_{n-\Delta}\}$ be a basis for $V^\perp$. We think of F and G as polynomials in $\mathbb{C}[v_1, \ldots, v_\Delta, u_1, \ldots, u_{n-\Delta}]$. As $T_{\boldsymbol{\alpha},V} : \mathbb{C}[v_1, \ldots, v_\Delta, u_1, \ldots, u_{n-\Delta}] \to \mathbb{C}[z, u_1, \ldots, u_{n-\Delta}]$, Theorem 17 implies that if $T_{\boldsymbol{\alpha},V}(F)$ and $T_{\boldsymbol{\alpha},V}(G)$ share a common factor that is not a polynomial in z, then, without loss of generality, their resultant with respect to u_1 is zero. Theorem 17 also implies that the resultant of F and G with respect to u_1 is not zero. Observe that with probability 1 over the choice of $\boldsymbol{\alpha}$, we have that $\deg_{u_1}(F) = \deg_{u_1}(T_{\boldsymbol{\alpha},V}(F))$ and $\deg_{u_1}(G) = \deg_{u_1}(T_{\boldsymbol{\alpha},V}(G))$. As $T_{\boldsymbol{\alpha},V}$ is a ring homomorphism this implies that $Res_{u_1}(T_{\boldsymbol{\alpha},V}(G), T_{\boldsymbol{\alpha},V}(F)) = T_{\boldsymbol{\alpha},V}(Res_{u_1}(G,F))$. The Schwartz-Zippel-DeMillo-Lipton lemma now implies that sending each basis element of V to a random multiple of z, chosen uniformly from $(0,1)$ will keep the resultant non zero with probability 1. This also means that $T_{\boldsymbol{\alpha},V}(F)$ and $T_{\boldsymbol{\alpha},V}(G)$ share no common factor. □

Corollary 30. *Let V be a Δ-dimensional linear space of linear forms. Let F and G be two linearly independent, irreducible quadratics, such that $Lin(F), Lin(G) \not\subseteq V$. Then, with probability 1 over the choice of $\boldsymbol{\alpha} \in [0,1]^\Delta$ (say according to the uniform distribution), $T_{\boldsymbol{\alpha},V}(F)$ and $T_{\boldsymbol{\alpha},V}(G)$ are linearly independent.*

Proof. As F and G are irreducible they share no common factors. Theorem 29 implies that $T_{\boldsymbol{\alpha},V}(F)$ and $T_{\boldsymbol{\alpha},V}(G)$ do not share a common factor that is not a polynomial in z. The Schwartz-Zippel-DeMillo-Lipton implies that with probability 1, $T_{\boldsymbol{\alpha},V}(F)$ and $T_{\boldsymbol{\alpha},V}(G)$ are not polynomials in z, and therefore they are linearly independent. □

Claim 31. *Let Q be an irreducible quadratic polynomial, and V a Δ-dimensional linear space. Then for every $\boldsymbol{\alpha} \in \mathbb{C}^\Delta$, $\text{rank}_s(T_{\boldsymbol{\alpha},V}(Q)) \geq \text{rank}_s(Q) - \Delta$.*

Proof. $\text{rank}_s(T_{\boldsymbol{\alpha},V}(Q)) \geq \text{rank}_s(T_{\boldsymbol{\alpha},V}(Q)|_{z=0}) = \text{rank}_s(Q|_{V=0}) \geq \text{rank}_s(Q) - \Delta$, where the last inequality follows from Theorem 22. □

Claim 32. *Let $\mathcal{Q}$ be a set of quadratics, and V be a Δ-dimensional linear space. Then, if there are linearly independent vectors, $\{\boldsymbol{\alpha}^1, \ldots, \boldsymbol{\alpha}^\Delta\} \subset \mathbb{C}^\Delta$, such that, for every i,*[3] $\dim(\mathit{Lin}(T_{\boldsymbol{\alpha}^i,V}(\mathcal{Q}))) \leq \sigma$ *then* $\dim(\mathit{Lin}(\mathcal{Q})) \leq (\sigma+1)\Delta$.

Proof. As $\dim(\text{Lin}(T_{\boldsymbol{\alpha}^i,V}(\mathcal{Q}))) \leq \sigma$, there are ${u^i}_1, \ldots, {u^i}_\sigma \subset V^\perp$ such that $\text{Lin}(T_{\boldsymbol{\alpha}^i,V}(\mathcal{Q})) \subseteq \text{span}\{z, {u^i}_1, \ldots, {u^i}_\sigma\}$. We will show that $\text{Lin}(\mathcal{Q}) \subset V + \text{span}\{\{{u^i}_1, \ldots, {u^i}_\sigma\}_{i=1}^\Delta\}$, which is of dimension at most $\Delta + \sigma\Delta$.

Let $P \in \mathcal{Q}$, then there are linear forms, $a_1, \ldots, a_\Delta \subset V^\perp$ and polynomials $P_V \in \mathbb{C}[V]$ and $P' \in \mathbb{C}[V^\perp]$, such that

$$P = P_V + \sum_{j=1}^{\Delta} a_j v_j + P'.$$

Therefore, after taking the projection for a specific $T_{\boldsymbol{\alpha}^i,V}$, for some $\gamma \in \mathbb{C}$,

$$T_{\boldsymbol{\alpha}^i,V}(P) = \gamma z^2 + \left(\sum_{j=1}^{\Delta} \alpha_j^i a_j\right) z + P'.$$

Denote $b_{P,i} = \sum_{j=1}^{\Delta} \alpha_j^i a_j$. By Theorem 30 if $a_1, \ldots, a_\Delta$ are not all zeros, then, with probability 1, $b_{P,i} \neq \mathbf{0}$.

If $b_{P,i} \notin \text{Lin}(P')$ then from Theorem 23 it follows that $\{z, b_{P,i}, \text{Lin}(P')\} \subseteq \text{span}\{\text{Lin}(T_{\boldsymbol{\alpha}^i,V}(P))\}$. If, on the other hand, $b_{P,i} \in \text{Lin}(P')$, then clearly $\{b_{P,i}, \text{Lin}(P')\} \subseteq \text{span}\{z, \text{Lin}(T_{\boldsymbol{\alpha}^i,V}(P))\}$. To conclude, in either case, $\{b_{P,i}, \text{Lin}(P')\} \subseteq \text{span}\{z, {u^i}_1, \ldots, {u^i}_\sigma\}$.

Applying the analysis above to $T_{\boldsymbol{\alpha}^1,V}, \ldots, T_{\boldsymbol{\alpha}^\Delta,V}$ we obtain that $\text{span}\{b_{P,1}, \cdots b_{P,\Delta}\} \subseteq \text{span}\{\{{u^i}_1, \ldots, {u^i}_\sigma\}_{i=1}^\Delta\}$. As $\boldsymbol{\alpha}^1, \ldots \boldsymbol{\alpha}^\Delta$ are linearly independent, we have that $\{a_1, \ldots, a_\Delta\} \subset \text{span}\{b_{P,1}, \cdots b_{P,\Delta}\}$, and thus $\text{Lin}(P) \subseteq V + \{a_1, \ldots, a_\Delta\} + LS(P') \subseteq V + \text{span}\{\{{u^i}_1, \ldots, {u^i}_\sigma\}_{i=1}^\Delta\}$. □

3 Sylvester-Gallai theorem for quadratic polynomials

In this section we prove Theorem 4. For convenience we repeat the statement of the theorem.

Theorem (Theorem 4)**.** *There exists a universal constant c such that the following holds. Let $\tilde{\mathcal{Q}} = \{Q_i\}_{i\in\{1,\ldots,m\}} \subset \mathbb{C}[x_1, \ldots, x_n]$ be a finite set of pairwise linearly independent homogeneous polynomials, such that every $Q_i \in \tilde{\mathcal{Q}}$ is either irreducible or a square of a linear form. Assume that, for every $i \neq j$, whenever Q_i and Q_j vanish then so does $\prod_{k\in\{1,\ldots,m\}\setminus\{i,j\}} Q_k$. Then,* $\dim(\text{span}\{\mathcal{Q}\}) \leq c$.

Remark 33. *The requirement that the polynomials are homogeneous is not essential as homogenization does not affect the property $Q_k \in \sqrt{\langle Q_i, Q_j\rangle}$.*

[3] Recall that $\text{Lin}(T_{\boldsymbol{\alpha}^i,V}(\mathcal{Q}))$ is the space spanned by $\cup_{Q\in\mathcal{Q}}\text{Lin}(T_{\boldsymbol{\alpha}^i,V}(\mathcal{Q}))$.

Remark 34. *Note that we no longer demand that the polynomials are irreducible but rather allow some of them to be square of linear forms, but now we restrict all polynomials to be of degree exactly* 2*. Note that both versions of the theorem are equivalent, as this modification does not affect the vanishing condition.*

We use the following claim of [17].

Claim 35 (Claim 11 in [17])**.** *Let* $P_1, \ldots, P_d, Q_1, \ldots, Q_k \in \mathbb{C}[x_1, \ldots, x_n]$ *be homogeneous and the degree of each* P_i *is at most* r*. Then,*

$$\prod_{i=1}^{k} Q_i \in \sqrt{\langle P_1, \ldots, P_d \rangle} \Rightarrow \exists \{i_1, \ldots, i_{r^d}\} \subset [k] \text{ such that } \prod_{j=1}^{r^d} Q_{i_j} \in \sqrt{\langle P_1, \ldots, P_d \rangle}\,.$$

Remark 36. *Note that from Theorem 35 it follows that for every* $i \neq j$ *there exists a subset* $\mathcal{K} \subseteq [m] \setminus \{i, j\}$ *such that* $|\mathcal{K}| \leq 4$ *and whenever* Q_i *and* Q_j *vanish then so does* $\prod_{k \in \mathcal{K}} Q_k$*.*

In what follows we shall use the following terminology. Whenever we say that two quadratics $Q_1, Q_2 \in \tilde{\mathcal{Q}}$ satisfy Theorem 5(i) we mean that there is a polynomial $Q_3 \in \tilde{\mathcal{Q}} \setminus \{Q_1, Q_2\}$ in their linear span. Similarly, when we say that they satisfy Theorem 5(ii) (Theorem 5(iii)) we mean that there is a reducible quadratic in their linear span (they belong to $\langle a_1, a_2 \rangle$ for linear forms a_1, a_2).

Proof of Theorem 4. Partition the polynomials to two sets. Let $\mathcal{L}$ be the set of all squares and let $\mathcal{Q}$ be the subset of irreducible quadratics, thus $\tilde{\mathcal{Q}} = \mathcal{Q} \cup \mathcal{L}$. Denote $|\mathcal{Q}| = m$, $|\mathcal{L}| = r$. Let $\delta = \frac{1}{100}$,
and denote

- $\mathcal{P}_1 = \{P \in \mathcal{Q} \mid$ There are at least δm polynomials in $\mathcal{Q}$ such that P satisfies Theorem 5(i) but not Theorem 5(ii) with each of them$\}$.
- $\mathcal{P}_3 = \{P \in \mathcal{Q} \mid$ There are at least δm polynomials in $\mathcal{Q}$ such that P satisfies Theorem 5(iii) with each of them$\}$.

The proof first deals with the case where $\mathcal{Q} = \mathcal{P}_1 \cup \mathcal{P}_3$. We then handle the case that there is $Q \in \mathcal{Q} \setminus (\mathcal{P}_1 \cup \mathcal{P}_3)$.

3.1 The case $\mathcal{Q} = \mathcal{P}_1 \cup \mathcal{P}_3$.

Assume that $\mathcal{Q} = \mathcal{P}_1 \cup \mathcal{P}_3$. For our purposes, we may further assume that $\mathcal{P}_1 \cap \mathcal{P}_3 = \emptyset$, by letting $\mathcal{P}_1 = \mathcal{P}_1 \setminus \mathcal{P}_3$.

Claim 37. *There exists a linear space of linear forms,* V*, such that* $\dim(V) = O(1)$ *and* $\mathcal{P}_3 \subset \langle V \rangle$*.*

The intuition behind the claim is based on the following observation.

Observation 38. *If* $Q_1, Q_2 \in \mathcal{Q}$ *satisfy Theorem 5(iii) then* $\dim(Lin(Q_1)), \dim(Lin(Q_2)) \leq 4$ *and* $\dim(Lin(Q_1) \cap Lin(Q_2)) \geq 2$*.*

Thus, we have many small dimensional spaces that have large pairwise intersections and we can therefore expect that such a V may exist.

Proof. We prove the existence of V by explicitly constructing it. Repeat the following process: Set $V = \{\mathbf{0}\}$, and $\mathcal{P}_3' = \emptyset$. At each step consider any $Q \in \mathcal{P}_3$ such that $Q \notin \langle V \rangle$ and set $V = \mathrm{Lin}(Q) + V$, and $\mathcal{P}_3' = \mathcal{P}_3' \cup \{Q\}$. Repeat this process as long as possible, i.e, as long as $\mathcal{P}_3 \not\subseteq \langle V \rangle$. We show next that this process must end after at most $\frac{3}{\delta}$ steps. In particular, $|\mathcal{P}_3'| \leq \frac{3}{\delta}$. It is clear that at the end of the process it holds that $\mathcal{P}_3 \subset \langle V \rangle$.

Claim 39. *Let $Q \in \mathcal{Q}$ and $\mathcal{B} \subseteq \mathcal{P}_3'$ be the subset of all polynomials in $\mathcal{P}_3'$ that satisfy Theorem 5(iii) with $\mathcal{Q}$, then $|\mathcal{B}| \leq 3$.*

Proof. Assume towards a contradiction that $|\mathcal{B}| \geq 4$, and that Q_1, Q_2, Q_3 and Q_4 are the first 4 elements of $\mathcal{B}$ that where added to $\mathcal{P}_3'$. Denote $U = \mathrm{Lin}(Q)$, and $U_i = U \cap \mathrm{Lin}(Q_i)$, for $1 \leq i \leq 4$.

As Q satisfies Theorem 5(iii) we have that $\dim(U) \leq 4$. Furthermore, for every i, $\dim(U_i) \geq 2$ (by Theorem 38). As the Q_is were picked by the iterative process, we have that $U_2 \not\subseteq U_1$. Indeed, since $Q_2 \in \langle U_2 \rangle$, if we had $U_2 \subseteq U_1 \subseteq \mathrm{Lin}(Q_1) \subseteq V$, then this would imply that $Q_2 \in \langle V \rangle$, in contradiction to the fact that $Q_2 \in \mathcal{P}_3'$. Similarly we get that $U_3 \not\subseteq U_1 + U_2$ and $U_4 \not\subseteq U_1 + U_3 + U_3$. However, as the next simple lemma shows, this is not possible.

Lemma 40. *Let V be a linear space of dimension ≤ 4, and let $V_1, V_2, V_3 \subset V$ each of dimension ≥ 2, such that $V_1 \not\subseteq V_2$ and $V_3 \not\subseteq V_2 + V_1$ then $V = V_1 + V_2 + V_3$.*

Proof. As $V_1 \not\subseteq V_2$ we have that $\dim(V_1 + V_2) \geq 3$. Similarly we get $4 \leq \dim(V_1 + V_2 + V_3) \leq \dim(V) = 4$. □

Thus, Theorem 40 implies that $V = U_1 + U_2 + U_3$ and in particular, $U_4 \subseteq U_1 + U_2 + U_3$ in contradiction. This completes the proof of Theorem 39. □

For $Q_i \in \mathcal{P}_3'$, define $T_i = \{Q \in \mathcal{Q} \mid Q, Q_i \text{ satisfiy Theorem 5(iii)}\}$. Since $|T_i| \geq \delta m$, and as by Theorem 39 each $Q \in \mathcal{Q}$ belongs to at most 3 different sets, it follows by double counting that $|\mathcal{P}_3'| \leq 3/\delta$. As in each step we add at most 4 linearly independent linear forms to V, we obtain $\dim(V) \leq \frac{12}{\delta}$.

This completes the proof of Theorem 37. □

So far V satisfies that $\mathcal{P}_3 \subset \langle V \rangle$. Next, we find a small set of polynomials $\mathcal{I}$ such that $\mathcal{Q} \subset \langle V \rangle + \mathrm{span}\{\mathcal{I}\}$.

Claim 41. *There exists a set $\mathcal{I} \subset \mathcal{Q}$ such that $\mathcal{Q} \subset \langle V \rangle + \mathrm{span}\{\mathcal{I}\}$ and $|\mathcal{I}| = O(1/\delta)$.*

Proof. As before the proof shows how to construct $\mathcal{I}$ by an iterative process. Set $\mathcal{I} = \emptyset$ and $\mathcal{B} = \mathcal{P}_3$. First add to $\mathcal{B}$ any polynomial from $\mathcal{P}_1$ that is in $\langle V \rangle$. Observe that at this point we have that $\mathcal{B} \subset \mathcal{Q} \cap \langle V \rangle$. We now describe another iterative process for the polynomials in $\mathcal{P}_1$. In each step pick any $P \in \mathcal{P}_1 \setminus \mathcal{B}$ such that P satisfies Theorem 5(i), but not Theorem 5(ii),[4] with at least $\frac{\delta}{3}m$ polynomials in $\mathcal{B}$, and add it to both $\mathcal{I}$ and to $\mathcal{B}$. Then, we add to $\mathcal{B}$ all the polynomials $P' \in \mathcal{P}_1$ that satisfy $P' \in \mathrm{span}\{(\mathcal{Q} \cap \langle V \rangle) \cup \mathcal{I}\}$. Note, that we always maintain that $\mathcal{B} \subset \mathrm{span}\{(\mathcal{Q} \cap \langle V \rangle) \cup \mathcal{I}\}$.

We continue this process as long as we can. Next, we prove that at the end of the process we have that $|\mathcal{I}| \leq 3/\delta$.

[4]By this we mean that there are many polynomials that together with P span another polynomial in $\mathcal{Q}$ but not in $\mathcal{L}$.

Claim 42. *In each step we added to $\mathcal{B}$ at least $\frac{\delta}{3}m$ new polynomials from $\mathcal{P}_1$. In particular, $|\mathcal{I}| \leq 3/\delta$.*

Proof. Consider what happens when we add some polynomial P to $\mathcal{I}$. By the description of our process, P satisfies Theorem 5(i) with at least $\frac{\delta}{3}m$ polynomials in $\mathcal{B}$. Any $Q \in \mathcal{B}$, that satisfies Theorem 5(i) with P, must span with P a polynomial $P' \in \tilde{\mathcal{Q}}$. Observe that $P' \notin \mathcal{L}$ as Q, P do not satisfy Theorem 5(ii), and thus $P' \in \mathcal{Q}$. It follows that $P' \in \mathcal{P}_1$ since otherwise we would have that $P \in \text{span}\{\mathcal{B}\} \subset \text{span}\{(\mathcal{Q} \cap \langle V \rangle) \cup \mathcal{I}\}$, which implies $P \in \mathcal{B}$ in contradiction to the way that we defined the process. Furthermore, for each such $Q \in \mathcal{B}$ the polynomial P' is unique. Indeed, if there was a $P \neq P' \in \mathcal{P}_1$ and $Q_1, Q_2 \in \mathcal{B}$ such that $P' \in \text{span}\{Q_1, P\} \cap \text{span}\{Q_2, P\}$ then by pairwise independence we would conclude that $P \in \text{span}\{Q_1, Q_2\} \subset \text{span}\{\mathcal{B}\}$, which, as we already showed, implies $P \in \mathcal{B}$ in contradiction. Thus, when we add P to $\mathcal{I}$ we add at least $\frac{\delta}{3}m$ polynomials to $\mathcal{B}$. In particular, the process terminates after at most $3/\delta$ steps and thus $|\mathcal{I}| \leq 3/\delta$. □

Consider the polynomials left in $\mathcal{P}_1 \setminus \mathcal{B}$. As they "survived" the process, each of them satisfies the condition in the definition of $\mathcal{P}_1$ with at most $\frac{\delta}{3}m$ polynomials in $\mathcal{B}$. From the fact that $\mathcal{P}_3 \subseteq \mathcal{B}$ and the uniqueness property we obtained in the proof of Theorem 42, we get that $\mathcal{P}_1 \setminus \mathcal{B}$ satisfies the conditions of Theorem 12 with parameter $\delta/3$ and thus, Theorem 13 implies that $\dim(\mathcal{P}_1 \setminus \mathcal{B}) \leq O(1/\delta)$. Adding a basis of $\mathcal{P}_1 \setminus \mathcal{B}$ to $\mathcal{I}$ we get that $|\mathcal{I}| = O(1/\delta)$ and every polynomial in $\mathcal{Q}$ is in $\text{span}\{(\mathcal{Q} \cap \langle V \rangle) \cup \mathcal{I}\}$. □

We are not done yet as the dimension of $\langle V \rangle$, as a vector space, is not a constant. Nevertheless, we next show how to use Sylvester-Gallai theorem to bound the dimension of $\mathcal{Q}$ given that $\mathcal{Q} \subset \text{span}\{(\mathcal{Q} \cap \langle V \rangle) \cup \mathcal{I}\}$. To achieve this we introduce yet another iterative process: For each $P \in \mathcal{Q} \setminus \langle V \rangle$, if there is quadratic L, with $\text{rank}_s(L) \leq 2$, such that $P + L \in \langle V \rangle$, then we set $V = V + \text{Lin}(L)$ (this increases the dimension of V by at most 4). Since this operation increases $\dim(\langle V \rangle \cap \mathcal{Q})$ we can remove one polynomial from $\mathcal{I}$, and thus decrease its size by 1, and still maintain the property that $\mathcal{Q} \subset \text{span}\{(\mathcal{Q} \cap \langle V \rangle) \cup \mathcal{I}\}$. We repeat this process until either $\mathcal{I}$ is empty, or none of the polynomials in $\mathcal{I}$ satisfies the condition of the process. By the upper bound on $|\mathcal{I}|$ the dimension of V grew by at most $4|\mathcal{I}| = O(1/\delta)$ and thus it remains of dimension $O(1/\delta) = O(1)$. At the end of the process we have that $\mathcal{Q} \subset \text{span}\{(\mathcal{Q} \cap \langle V \rangle) \cup \mathcal{I}\}$ and that every polynomial in $P \in \mathcal{Q} \setminus \langle V \rangle$ has $\text{rank}_s(P) > 2$, even if we set all linear forms in V to zero.

Consider the map $T_{\boldsymbol{\alpha},V}$ as given in Theorem 27, for a randomly chosen $\boldsymbol{\alpha} \in [0,1]^{\dim(V)}$. Each polynomial in $\mathcal{Q} \cap \langle V \rangle$ is mapped to a polynomial of the form form zb, for some linear form b. From Theorem 22, it follows that every polynomial in $\mathcal{Q} \setminus \langle V \rangle$ still has rank larger than 2 after the mapping. Let

$$\mathcal{A} = \{b \mid \text{ some polynomial in } \mathcal{Q} \cap \langle V \rangle \text{ was mapped to } zb\} \cup T_{\boldsymbol{\alpha},V}(\mathcal{L}) \,.$$

We now show that, modulo z, $\mathcal{A}$ satisfies the conditions of Sylvester-Gallai theorem. Let $b_1, b_2 \in \mathcal{A}$ such that $b_1 \notin \text{span}\{z\}$ and $b_2 \notin \text{span}\{z, b_1\}$. As $\tilde{\mathcal{Q}}$ satisfies the conditions of Theorem 4 we get that there are polynomials $Q_1, \ldots, Q_4 \in \tilde{\mathcal{Q}}$ such that $\prod_{i=1}^{4} T_{\boldsymbol{\alpha},V}(Q_i) \in \sqrt{\langle b_1, b_2 \rangle} = \langle b_1, b_2 \rangle$, where the equality holds as

$\langle b_1, b_2\rangle$ is a prime ideal. This fact also implies that, without loss of generality, $T_{\boldsymbol{\alpha},V}(Q_4) \in \langle b_1, b_2\rangle$. Thus, $T_{\boldsymbol{\alpha},V}(Q_4)$ has rank at most 2 and therefore $Q_4 \in \mathcal{L} \cup (\mathcal{Q} \cap \langle V\rangle)$. Hence, $T_{\boldsymbol{\alpha},V}(Q_4)$ was mapped to zb_4 or to b_4^2. In particular, $b_4 \in \mathcal{A}$. Theorem 29 and Theorem 30 imply that b_4 is neither a multiple of b_1 nor a multiple of b_2, so it must hold that b_4 depends non-trivially on both b_1 and b_2. Thus, $\mathcal{A}$ satisfies the conditions of Sylvester-Gallai theorem modulo z. It follows that $\dim(\mathcal{A}) = O(1)$.

The argument above shows that the dimension of $T_{\boldsymbol{\alpha},V}(\mathcal{L}\cup(\mathcal{Q}\cap\langle V\rangle)) = O(1)$. Theorem 32 implies that if we denote $U = \text{span}\{\mathcal{L} \cup \text{Lin}(\mathcal{Q} \cap \langle V\rangle)\}$ then $\dim(U)$ is $O(1)$. As $\mathcal{Q} \subseteq \text{span}\{(\mathcal{Q} \cap \langle V\rangle) \cup \mathcal{I}\}$, we obtain that $\dim(\tilde{\mathcal{Q}}) = \dim(\mathcal{L} \cup \mathcal{Q}) = O(1)$, as we wanted to show.

This completes the proof of Theorem 4 for the case $\mathcal{Q} = \mathcal{P}_1 \cup \mathcal{P}_3$.

3.2 The case $\mathcal{Q} \neq \mathcal{P}_1 \cup \mathcal{P}_3$.

In this case there is some polynomial $Q_o \in \mathcal{Q} \setminus (\mathcal{P}_1 \cup \mathcal{P}_3)$. In particular, Q_0 satisfies Theorem 5(ii) with at least $(1 - 2\delta)m$ of the polynomials in $\mathcal{Q}$; of the remaining polynomials, at most δm satisfy Theorem 5(i) with Q_o; and, Q_o satisfies Theorem 5(iii) with at most δm polynomials. Let

- $\mathcal{Q}_1 = \{P \in \mathcal{Q} \mid P, Q_o \text{ satisfiy Theorem 5(ii) }\} \cup \{Q_o\}$
- $\mathcal{Q}_2 = \{P \in \mathcal{Q} \mid P, Q_o \text{ do not satisfiy Theorem 5(ii) }\}$
- $m_1 = |\mathcal{Q}_1|$, $m_2 = |\mathcal{Q}_2|$.

As $Q_o \notin \mathcal{P}_1 \cup \mathcal{P}_3$ we have that $m_2 \leq 2\delta m$ and $m_1 \geq (1 - 2\delta)m$.

These properties of Q_o and $\mathcal{Q}$ are captured by the following definition.

Definition 43. *Let $\mathcal{Q}_1 = \{Q_o, Q_1, \ldots, Q_{m_1}\}$ and $\mathcal{Q}_2 = \{P_1, \ldots, P_{m_2}\}$ be sets of irreducible homogeneous quadratic polynomials. Let $\mathcal{L} = \{{\ell^2}_1, \ldots, {\ell^2}_r\}$ be a set of squares of homogeneous linear forms. We say that $\tilde{\mathcal{Q}} = \mathcal{Q} \cup \mathcal{L}$ where $\mathcal{Q} = \mathcal{Q}_1 \cup \mathcal{Q}_2$ is a (Q_o, m_1, m_2)-set if it satisfies the following:*

1. *$\tilde{\mathcal{Q}}$ satisfy the conditions in the statement of Theorem 4.*
2. *$m_1 > 5m_2 + 2$.*
3. *For every $j \in [m_1]$, there are linear forms a_j, b_j such that $Q_j = Q_o + a_j b_j$.*
4. *For every $i \in [m_2]$, every non-trivial linear combination of P_i and Q_o has rank at least 2.*
5. *At most m_2 of the polynomials in $\mathcal{Q}$ satisfy Theorem 5(iii) with Q_o.*

By the discussion above, the following theorem is what we need in order to complete the proof for the case $\mathcal{Q} \neq \mathcal{P}_1 \cup \mathcal{P}_3$.

Theorem 44. *Let $\tilde{\mathcal{Q}}$ satisfy the conditions of Theorem 43, then* $\dim \tilde{\mathcal{Q}} = O(1)$.

We prove this theorem in section 4. This concludes the proof of Theorem 17. □

4 Proof of Theorem 44

In this section we prove Theorem 44. The proof is divided to two parts according to whether the polynomial Q_o in Theorem 43 is of high rank (Theorem 46) or of low rank (Theorem 60). Each part is also divided to two – first we consider what happens when $m_2 = 0$ and then the general case where $m_2 \neq 0$. The reason for this split is that when Q_o is of high rank then we know, e.g., that it cannot satisfy Theorem 5(iii) with any other polynomial. Similarly any polynomial satisfying Theorem 5(ii) with Q_o is also of high rank and cannot satisfy Theorem 5(iii) with any other polynomial. The reason why we further break the argument to weather $m_2 = 0$ or not, is that when $m_2 = 0$ all the polynomials are of the form $Q_o + ab$ for some linear forms a, b, which means we have fewer cases to analyse.

While this seems a bit restrictive, the general case is not much harder and most of the ideas there already appear in the case $m_2 = 0$.

Throughout the proof we use the notation of Theorem 43. In particular, each $Q_i \in \mathcal{Q}_1$ is of the form $Q_i = Q_o + a_i b_i$.

4.1 Q_o is of high rank

In this subsection we assume that $\tilde{\mathcal{Q}}$ is a (Q_o, m_1, m_2)-set for some quadratic Q_o of rank at least 100, this constant is arbitrary, as we just need it to be large enough. The following observation says that for our set $\mathcal{Q}$ we will never have to consider Theorem 5(iii).

Observation 45. *For $\tilde{\mathcal{Q}} = \mathcal{Q} \cup \mathcal{L}$ that satisfy Theorem 43 with $\mathrm{rank}_s(Q_o) \geq 100$, for every $j \in [m_1]$ the rank of Q_j is at least $100 - 1 > 2$ and so Q_j never satisfies Theorem 5(iii) with any other polynomial in $\tilde{\mathcal{Q}}$.*

Our goal in this subsection is to prove the next claim.

Claim 46. *Let $\tilde{\mathcal{Q}} = \mathcal{Q} \cup \mathcal{L}$ be a (Q_o, m_1, m_2)-set with $\mathrm{rank}_s(Q_o) \geq 100$. Then $\dim(\mathrm{span}\{\tilde{\mathcal{Q}}\}) = O(1)$.*

We break the proof of Theorem 46 to two steps. First we handle the case $m_2 = 0$ and then the case $m_2 \neq 0$.

4.1.1 The case $m_2 = 0$

In this subsection we prove the following version of Theorem 46 for the case $m_2 = 0$.

Claim 47. *Let $\tilde{\mathcal{Q}} = \mathcal{Q} \cup \mathcal{L}$ be a $(Q_o, m_1, 0)$-set with $\mathrm{rank}_s(Q_o) \geq 100$. Then, for a_i, b_i, ℓ_j as in Theorem 43, $\dim(\mathrm{span}\{a_1, \ldots, a_{m_1}, b_1, \ldots, b_{m_1}, \ell_1, \ldots, \ell_r\}) \leq 7$. In particular, $\dim(\mathrm{span}\{\mathcal{Q}\}) \leq 8$.*

We first show some properties satisfied by the products $\{a_1 b_1, \ldots, a_{m_1} b_{m_1}\}$.

Remark 48. *For $\ell_i^2 \in \mathcal{L}$ we can write $\ell_i^2 = 0 \cdot Q_o + \ell_i \ell_i$. Thus, from now on we can assume that every $Q_i \in \tilde{\mathcal{Q}}$ is of the form $Q_i = \alpha_i Q_o + a_i b_i$, for $\alpha_i \in \{0, 1\}$, and when $\alpha_i = 0$ it holds that $a_i = b_i$. We shall use the convention that for $i \in \{m_1 + 1, \ldots, m_1 + r\}$, $a_i = \ell_{i - m_1}$.*

Claim 49. *Let $\tilde{\mathcal{Q}} = \mathcal{Q} \cup \mathcal{L}$ be a $(Q_o, m_1, 0)$-set with $\mathrm{rank}_s(Q_o) \geq 100$, and let $Q_i = Q_o + a_i b_i$ and $Q_j = Q_o + a_j b_j$ be polynomials in $\mathcal{Q} = \mathcal{Q}_1$.*

1. *If Q_i and Q_j satisfy Theorem 5(i) then there exists $k \in [m_1 + r]$ such that for some $\alpha, \beta \in \mathbb{C} \setminus \{0\}$*

$$\alpha a_i b_i + \beta a_j b_j = a_k b_k. \tag{1}$$

2. *If Q_i and Q_j satisfy Theorem 5(ii) then there exist two linear forms, c and d such that*

$$a_i b_i - a_j b_j = cd. \tag{2}$$

The claim only considers Theorem 5(i) and Theorem 5(ii) as by Theorem 45 we know that Q_i, Q_j do not satisfy Theorem 5(iii). Note that the guarantee of this claim is not sufficient to conclude that the dimension of $a_1, \ldots, a_{m_1}, b_1, \ldots, b_{m_1}$ is bounded. The reason is that c and d are not necessarily part of the set. For example if for every i, $a_i b_i = x_i^2 - x_1^2$. Then every pair, Q_i, Q_j satisfy Theorem 5(ii), but the dimension of $a_1, \ldots, a_{m_1}, b_1, \ldots, b_{m_1}$ is unbounded.

Proof of Theorem 49. If Q_i, Q_j satisfy Theorem 5(i) then there are constants $\alpha, \beta \in \mathbb{C}$ and $k \in [m_1 + r] \setminus \{i, j\}$ such that $\alpha(Q_o + a_i b_i) + \beta(Q_o + a_j b_j) = \alpha Q_i + \beta Q_j = Q_k = \alpha_k Q_o + a_k b_k$. Rearranging we get that

$$\alpha a_i b_i + \beta a_j b_j - a_k b_k = (\alpha_k - (\alpha + \beta)) Q_o \, .$$

From the fact that $\mathrm{rank}_s(Q_o) \geq 100$, it must be that $\alpha_k - (\alpha + \beta) = 0$. Hence,

$$\alpha a_i b_i + \beta a_j b_j = a_k b_k \tag{3}$$

and (1) holds. Observe that $\alpha, \beta \neq 0$ as otherwise we will have two linearly dependent polynomials in $\mathcal{Q}$.

If Q_i, Q_j satisfy Theorem 5(ii) then there are $\alpha, \beta \in \mathbb{C}$ and two linear forms c and d such that $\alpha(Q_o + a_i b_i) + \beta(Q_o + a_j b_j) = cd$, and again, by the same argument, we get that $\beta = -\alpha$, and that, without loss of generality,

$$a_i b_i - a_j b_j = cd. \qquad \square$$

Let $V_i := \mathrm{span}\{a_i, b_i\}$. We next show that the different spaces V_i satisfy some non-trivial intersection properties.

Claim 50. *Let $\tilde{\mathcal{Q}}$ be a $(Q_o, m_1, 0)$-set such that $\mathrm{rank}_s(Q_o) \geq 100$. If for some $i \in [m_1]$ we have $\dim(V_i) = 2$ then for every $j \in [m_1]$ it holds that $\dim(V_j \cap V_i) \geq 1$. In particular it follows that if $dim(V_j) = 1$ then $V_j \subsetneq V_i$.*

Proof. This follows immediately from Theorem 49 and Theorem 24. $\square$

Next we use this fact to conclude some structure on the set of pairs (a_i, b_i).

Claim 51. *Let $\tilde{\mathcal{Q}}$ be as in Theorem 47. If $\dim(\mathrm{span}\{a_i, b_i\}) > 3$ then there is a linear space of linear forms, V such that $\dim(V) \leq 4$, and for all $i \in [m_1 + r]$, $b_i \in \mathrm{span}\{a_i, V\}$ or $a_i \in \mathrm{span}\{b_i, V\}$.*

Proof. Consider the set of all V_i's of dimension 2. Combining Theorem 49 and Theorem 26 we get that either $\dim(\bigcup_{i=1}^m V_i) \leq 3$ or $\dim(\bigcap_{i=1}^m V_i) = 1$. If $\dim(\bigcup_{i=1}^m V_i) \leq 3$ then $V = \bigcup_{i=1}^m V_i$ is the linear space promised in the claim. If $\bigcap_{i=1}^m V_i) = 1$ there is a linear form, w, such that $\text{span}\{w\} = \dim(\bigcap_{i=1}^m V_i)$. It follows that for every $i \in [m_1]$ there are constants ε_i, δ_i such that, with out loss of generality, $b_i = \varepsilon_i a_i + \delta_i w$. Note that if $\dim(V_i) = 1$ this representation also holds with $\delta_i = 0$, and thus $V = \text{span}\{w\}$. is the linear space promised in the claim. □

From now on we assume there is a linear space of linear forms, V such that $\dim(V) \leq 4$ and for every $i \in [m_1 + r]$ it holds that $b_i = \varepsilon_i a_i + v_i$ (we can do this by replacing the roles of a_i and b_i if needed). Indeed, if $\dim(\text{span}\{a_i, b_i\}) > 3$ then this follows from Theorem 51 and otherwise we can take $V = \text{span}\{a_i, b_i\}$. Thus, following Theorem 48, every polynomial in $\mathcal{Q}$ is of the form $\alpha_i Q + a_i(\varepsilon_i a_i + v_i)$ and for polynomials in $\mathcal{L}$ we have that $\alpha_i = 0$, $\varepsilon_i = 1$ and $v_i = 0$.

The following claim is the crux of the proof of Theorem 47. It shows that, modulo V, the set $\{a_1, \ldots, a_{m_1+r}\}$ satisfies the Sylvester-Gallai theorem..

Claim 52. *Let $i \neq j \in [m_1 + r]$ be such that $a_i \notin V$ and $a_j \notin \text{span}\{a_i, V\}$. Then, there is $k \in [m_1 + r]$ such that $a_k \in \text{span}\{a_i, a_j, V\}$ and $a_k \notin \text{span}\{a_i, V\} \cup \text{span}\{a_j, V\}$.*

Proof. We split the proof to three cases (recall Theorem 48): Either (i) $\alpha_i = \alpha_j = 1$, or (ii) $\alpha_i = 1, \alpha_j = 0$ (without loss of generality), or (iii) $\alpha_i = \alpha_j = 0$. Recall that $\alpha_i = 0$ if and only if $i \in \{m + 1, \ldots, m + r\}$.

(i) $\alpha_i = \alpha_j = 1$. Theorem 49 implies that there are two linear forms c and d such that cd is a nontrivial linear combination of $a_j(\varepsilon_j a_j + v_j), a_i(\varepsilon_i a_i + v_i)$. We next show that without loss of generality c depends non-trivially on both a_i and a_j.

Lemma 53. *In the current settings, without lost of generality, $c = \mu a_i + \eta a_j$ where $\mu, \eta \neq 0$.*

Proof. Setting $a_i = 0$ gives that, without loss of generality, $cd \equiv_{a_i} a_j(\varepsilon_j a_j + v_j)$ and as $a_j \notin \text{span}\{a_i, V\}$ we have that $cd \not\equiv_{a_i} 0$. Thus, without loss of generality $c \equiv_{a_i} \eta a_j$, for some non-zero η. Let μ and η be such that $c = \mu a_i + \eta a_j$. We will now show that $\mu \neq 0$. Indeed, if this was not the case then we would have that $cd = \eta a_j d$. This means that $a_i(\varepsilon_i a_i + v_i) \in \text{span}\{a_j(\varepsilon_j a_j + v_j), \eta a_j d\}$ (since the linear dependence was non-trivial) setting $a_j = 0$ we see that either a_i, or $\varepsilon_i a_i + v_i$ in $\text{span}\{a_j\}$, which contradicts our assumption.

□

Equation 1 and Theorem 53 show that if Q_i and Q_j satisfy Theorem 5(i), i.e. they span Q_k (for $k \notin \{i, j\}$), then one of $a_k, \varepsilon_k a_k + v_k$ is a non-trivial linear combination of a_i and a_j. Thus, modulo V, a_k is in the span of a_i and a_j, which is what we wanted to show.

We next handle the case where Q_i and Q_j satisfy Theorem 5(ii). Let cd be a product of linear forms in the span of Q_i and Q_j. From Theorem 53

we can assume that $c = \mu a_i + \eta a_j$ with $\mu\eta \neq 0$. In particular, this means that $\sqrt{\langle Q_i, Q_j\rangle} = \sqrt{\langle cd, Q_j\rangle}$.

The assumption that $\mathrm{rank}_s(Q_o) \geq 100$ implies that Q_j is irreducible even after setting $c = 0$. It follows that if a product of irreducible polynomials satisfy $\prod_i A_i \in \sqrt{\langle cd, Q_j\rangle}$ then, after setting $c = 0$, some A_i is divisible by $Q_j|_{c=0}$. Thus, there is a multiplicand that is equal to $\alpha Q_j + ce$ for some linear form e. In particular, there must be a polynomial Q_k, $k \in [m_1 + r] \setminus \{i, j\}$, such that $Q_k = \alpha Q_j + ce$. If $\alpha = 0$ then it holds that $Q_k = a_k^2 = ce$ and therefore a_k satisfies the claim. Otherwise, as before, the rank condition on Q_o implies that $\alpha = 1$ and thus $a_k(\varepsilon_k a_k + v_k) = a_j(\varepsilon_j a_j + v_j) + (\mu a_i + \eta a_j)e$. Consider what happens when we set $a_j = 0$. We get that $a_k(\varepsilon_k a_k + v_k) \equiv_{a_j} \mu a_i e$. Note that it cannot be the case that $e \equiv_{a_j} 0$ as this would imply that $a_k \in \mathrm{span}\{a_j, v_k\}$ and in turn, this implies that $a_i \in \mathrm{span}\{a_j, V\}$ in contradiction to the choice of a_i and a_j. Thus, we get that either a_k or $\varepsilon_k a_k + v_k$ are equivalent to a_i modulo a_j. We next show that if either of them depends only on a_i, then we get a contradiction. Thus, we are left in the case that $a_k = \lambda a_i$ (the case $\varepsilon_k a_k + v_k = \lambda a_i$ is equivalent). Since $Q_k = Q_o + \lambda a_i\,(\varepsilon_k \lambda a_i + v_k) = Q_j + ce$ and we have that $Q_i = Q_o + a_i(\varepsilon_i a_i + v_i) = Q_j + cd$ we get by subtracting Q_i from Q_k that

$$a_i\left((\lambda^2\varepsilon_k - \varepsilon_i)a_i + (\lambda v_k - v_i)\right) = \lambda a_i(\varepsilon_k \lambda a_i + v_k) - a_i(\varepsilon_i a_i + v_i) = Q_k - Q_i = c(e-d)\ ,$$

and clearly neither side of the equation is zero since $Q_i \neq Q_k$. This implies that $c \in \mathrm{span}\{a_i, V\}$, in contradiction. Thus, in this case too we get that a_k satisfies the claim.

(ii) $\alpha_i = 1, \alpha_j = 0$. In this case, Q_i, Q_j must satisfy Theorem 5(ii), as $0 \cdot Q_i + Q_j = a_j^2$. As before, the assumption that $\mathrm{rank}_s(Q_o) \geq 100$ implies that Q_i is irreducible even after setting $a_j = 0$. It follows that if a product of irreducible polynomials satisfy $\prod_t A_t \in \sqrt{\langle a_j^2, Q_i\rangle}$ then, after setting $a_j = 0$, some A_t is divisible by $Q_i|_{a_j=0}$. In our case we get that there is a multiplicand that is equal to $\alpha Q_i + a_j e$ for some linear form e. In particular, there must be a polynomial Q_k, for $k \in [m_1 + r] \setminus \{i, j\}$, such that $Q_k = \alpha Q_i + a_j e$. If $\alpha = 0$ it follows that Q_k is reducible and thus of the form $Q_k = a_k^2 = a_j e$ which is a contradiction to pairwise linear independence (as $Q_k \sim Q_j$). Thus $\alpha = \alpha_k = 1$, and $a_k(\varepsilon_k a_k + v_k) = a_i(\varepsilon_i a_i + v_k) + a_j e$. As before, we can conclude that $a_k \in \mathrm{span}\{a_i, a_j, V\}$ and that it cannot be the case that $a_k \in \mathrm{span}\{a_i, V\} \cup \mathrm{span}\{a_j, V\}$ (as by rearranging the equation we will get a contradiction to the fact that $a_j \notin \mathrm{span}\{a_i, V\}$), which is what we wanted to show.

(iii) $\alpha_i = \alpha_j = 0$. Then $\sqrt{\langle Q_i, Q_j\rangle} = \langle a_i, a_j\rangle$ is a prime ideal. It follows that there is $k \in [m_1 + r] \setminus \{i, j\}$ such that $Q_k \in \langle a_i, a_j\rangle$ the rank condition on Q_o implies that $\alpha_k = 0$ and therefore a_k is a non-trivial linear combination of a_i and a_j, which is what we wanted to show.

This completes the proof of Theorem 52. □

We can now prove Theorem 47.

Proof of Theorem 47. Theorem 52 implies that any two linear functions in $\{a_1, \ldots, a_{m_1+r}\}$ that are linearly independent modulo V, span (modulo V) a third function in the set. This implies that if we project all the linear functions to the perpendicular space to V then they satisfy the usual condition of the Sylvester-Gallai theorem and thus the dimension of the projection is at most 3. As $\text{span}\{a_1, \ldots, a_{m_1}, b_1, \ldots, b_{m_1}, a_{m_1+1}, \ldots, a_{m_1+r}\} \subseteq \text{span}\{a_1, \ldots, a_{m_1+r}, V\}$, we get that $\dim(\{a_1, \ldots, a_{m_1}, b_1, \ldots, b_{m_1}, a_{m_1+1}, \ldots, a_{m_1+r}\}) \leq 3 + \dim(V) \leq 7$, as claimed. □

Thus far we have proved Theorem 47 which is a restriction of Theorem 46 to the case $m_2 = 0$. In the next subsection we handle the general case $m_2 \neq 0$.

4.1.2 The case $m_2 \neq 0$.

In this subsection we prove Theorem 46. We shall assume without loss of generality that $m_2 \neq 0$.

We first show that each $P_i \in \mathcal{Q}_2$ (recall Theorem 43) is either a rank-2 quadratic, or it is equal to Q_o plus a rank-2 quadratic.

Claim 54. *Let $\tilde{\mathcal{Q}}$ be a (Q_o, m_1, m_2)-set such that $\text{rank}_s(Q_o) \geq 100$. Then for every $i \in [m_2]$ there exists $\gamma_i \in \mathbb{C}$ such that $\text{rank}_s(P_i - \gamma_i Q_o)) = 2$.*

Proof. Fix $i \in [m_2]$. We shall analyse, for each $j \in [m_1]$, which case of Theorem 5 Q_j and P_i satisfy. From Theorem 45 we know that P_i does not satisfy Theorem 5(iii) with any Q_j. We start by analysing what happens when P_i and Q_j satisfy Theorem 5(ii). By definition, there exist linear forms a', b' and non zero constants $\alpha, \beta \in \mathbb{C}$, such that $\alpha P_i + \beta Q_j = a'b'$ and thus,

$$P_i = \frac{1}{\alpha}\left(a'b' - \beta\left(Q_o + a_j b_j\right)\right) = \frac{-\beta}{\alpha} Q_o + \left(\frac{1}{\alpha} a'b' - \frac{\beta}{\alpha} a_j b_j\right) . \tag{4}$$

Hence, the statement holds with $\gamma_i = -\frac{\beta}{\alpha}$. Indeed, observe that the rank_s of $(\frac{1}{\alpha} a'b' - \frac{\beta}{\alpha} a_j b_j)$ cannot be 1 as this will contradict item 4 in Theorem 43.

Thus, the only case left to consider is when P_i satisfies Theorem 5(i) alone with all the Q_j's. If for some $j \in [m_1]$ there is $j' \in [m_1]$ such that $Q_{j'} \in \text{span}\{Q_j, P_i\}$, then there are $\alpha, \beta \in \mathbb{C} \setminus \{0\}$, for which $P_i = \alpha Q_j + \beta Q_{j'}$ and then

$$P_i = (\alpha + \beta) Q_o + \alpha a_j b_j + \beta a_{j'} b_{j'},$$

and the statement holds with $\gamma_i = \beta + \alpha$. So, let us assume that for every $j \in [m_1]$, there is $t_j \in [m_2]$ such that $P_{t_j} \in \text{span}\{Q_j, P_i\}$. As $5m_2 + 2 < m_1$ there must be $j' \neq j'' \in [m_1]$ and $t' \in [m_2]$ such that $P_{t'} \in \text{span}\{Q_{j'}, P_i\}$ and $P_{t'} \in \text{span}\{Q_{j''}, P_i\}$. Since $\mathcal{Q}$ is a set of pairwise linearly independent polynomials, we can deduce that $\text{span}\{P_i, P_{t'}\} = \text{span}\{Q_{j'}, Q_{j''}\}$. In particular there exist $\alpha, \beta \in \mathbb{C}$, for which $P_i = \alpha Q_j + \beta Q_{j'}$, which, as we already showed, implies what we wanted to prove. □

For simplicity, rescale P_i so that $P_i = \gamma_i Q_o + L_i$ with $\text{rank}_s(L_i) = 2$ and $\gamma_i \in \{0, 1\}$. Clearly $\mathcal{Q}$ still satisfies the conditions of Theorem 43 after this rescaling, as it does not affect the vanishing conditions or linear independence. The next claim shows that even in the case $m_2 \neq 0$, the linear forms $\{a_1, \ldots, a_{m_1}, b_1, \ldots, b_{m_1}\}$ "mostly" belong to a low dimensional space (similar to Theorem 47).

Claim 55. *Let $\tilde{\mathcal{Q}}$ be a (Q_o, m_1, m_2)-set such that* $\text{rank}_s(Q_o) \geq 100$. *Then, there exists a subspace V of linear forms such that* $\dim(V) \leq 4$ *and that for at least $m_1 - m_2$ indices $j \in [m_1]$ it holds that $a_j, b_j \in V$. Furthermore, there is a polynomial $P \in \mathcal{Q}_2$ such that $P = \gamma Q_o + L$ and $Lin(L) = V$.*

Proof. Let $P_1 = \gamma_1 Q_o + L_1$ where $\text{rank}_s(L_1) = 2$. To simplify notation we drop the index 1 and only talk of P, L and γ. Set $V = \text{Lin}(L)$. As before, Theorem 45 implies that P cannot satisfy Theorem 5(iii) with any $Q_j \in \mathcal{Q}_1$.

Let $Q_j \in \mathcal{Q}_1 \cup \mathcal{L}$. If Q_j, P satisfy Theorem 5(iii), then $\alpha_j = 0$ and $Q_j = a_j^2$. By the rank condition on Q_o it follows that $\gamma = 0$ and therefore $a_j \in \text{Lin}(L) = V$.

Let $Q_j \in \mathcal{Q}_1 \cup \mathcal{L}$ be such that Q_j and P satisfy Theorem 5(ii). This means that there are two linear forms e, f, and non zero $\alpha, \beta \in \mathbb{C}$ for which $\alpha P - \beta Q_j = ef$, and so,

$$(\alpha\gamma - \beta\alpha_j)Q_o = -\alpha L + \beta a_j b_j + ef \tag{5}$$

As we assumed that $\text{rank}_s(Q_o) \geq 100$ this implies that $\alpha\gamma - \beta\alpha_j = 0$ and thus $\beta a_j b_j + ef = \beta L$. Theorem 19 implies that $e, f, a_j, b_j \in V$.

We have shown that V contains all a_j, b_j that come from polynomials satisfying Theorem 5(ii) with P.

Let $j \in [m_1]$ be such that P and Q_j satisfy Theorem 5(i) but not Theorem 5(ii), i.e, they span another polynomial in $\tilde{\mathcal{Q}} \setminus \mathcal{L}$. If this polynomial is in $\mathcal{Q}_1$, i.e. there exists $j' \in [m_1]$ such that $Q_{j'} \in \text{span}\{P, Q_j\}$ then $P = \alpha Q_j + \beta Q_{j'}$ and as before we would get that $a_{j'}, b_{j'}, a_j, b_j \in V$.

All that is left is to bound the number of $j \in [m_1]$ so that P and Q_j span a polynomial in $\mathcal{Q}_2$. If there are more than m_2 such indices j then, by the pigeonhole principle, for two of them, say j, j' it must be the case that there is some $i \in [m_2]$ such that $P_i \in \text{span}\{P, Q_j\}$ and $P_i \in \text{span}\{P, Q_{j'}\}$. As our polynomials are pairwise independent this implies that $P \in \text{span}\{Q_j, Q_{j'}\}$, and as before we get that $a_{j'}, b_{j'}, a_j, b_j \in V$.

It follows that the only j's for which we may have $a_j, b_j \notin V$ must be such that Q_j and P span a polynomial in $\mathcal{Q}_2$, and no other $Q_{j'}$ spans this polynomial with P. Therefore, there are at most m_2 such "bad" j's and the claim follows. □

Remark 56. *The proof of Theorem 55 implies that if $Q_i = \alpha_i Q_o + a_i b_i \in \mathcal{Q}_1$ satisfies that $\{a_i, b_i\} \not\subseteq V$ then it must be the case that Q_i and P span a polynomial $P_j \in \mathcal{Q}_2$.*

Claim 57. *Let $\tilde{\mathcal{Q}}$ be a (Q_o, m_1, m_2)-set such that* $\text{rank}_s(Q_o) \geq 100$. *Then there exists a 4-dimensional linear space V, such that for every $P_i \in \tilde{\mathcal{Q}}$ either P_i is defined over V, or there is a quadratic polynomial P_i' and a linear form v_i that are defined over V, and a linear form c_i, such that $P_i = Q_o + P_i' + c_i(\varepsilon_i c_i + v_i)$, or $P_i = c_i^2$.*

Proof. Theorem 55 implies the existence of a polynomial $P = \gamma Q_o + L \in \mathcal{Q}_2$ and 4-dimensional linear space $V = \text{Lin}(L)$ such that the set $\mathcal{I} = \{Q_j \mid j \in [m_1] \text{ and } a_j, b_j \in V\}$ satisfies $|\mathcal{I}| \geq m_1 - m_2$. We will prove that V is the space guaranteed in the claim. We first note that every $P_i \in \mathcal{I}$ satisfies the claim with $P_i' = a_i b_i$ and $v_i = c_i = 0$, and clearly for $Q_i \in \mathcal{L}$ the claim trivially holds.

Consider $Q_i \in \mathcal{Q}_1 \setminus \mathcal{I}$. By Theorem 56 it must be the case that Q_i and P span a polynomial $P_j \in \mathcal{Q}_2$. Hence, there are $\alpha, \beta \in \mathbb{C} \setminus \{0\}$ such that

$P_j = \alpha P + \beta Q_i$. From Theorem 54 we get that $P_j = \gamma_j Q_o + L_j$ and thus

$$(\gamma_j - \alpha\gamma - \beta)Q_o = \alpha L + \beta a_i b_i - L_j \ .$$

As $\text{rank}_s(Q_o) \geq 100$ it follows that $(\gamma_j - \alpha\gamma - \beta) = 0$ and $\alpha L + \beta a_i b_i = L_j$. Theorem 23 implies that $\text{span}\{a_i, b_i\} \cap V \neq \{\mathbf{0}\}$ and therefore there is $v_i \in V$ such that, without loss of generality, $b_i = \varepsilon_i a_i + v_i$, for some constant ε_i. Thus, the claimed statement holds for Q_i with $c_i = a_i$ and $Q'_i = 0$. I.e., $Q_i = Q_o + 0 + a_i(\varepsilon_i a_i + v_i)$.

Consider a polynomial $P_i = \gamma_i Q_o + L_i \in \mathcal{Q}_2$.

If $\gamma_i = 0$ then by rank argument we see that P_i cannot satisfy Theorem 5(ii) nor Theorem 5(iii) with any polynomial in $\mathcal{Q}_1$. Hence it must satisfy Theorem 5(i) with all the polynomials in $\mathcal{Q}_1$. Therefore, by the pigeonhole principle P_i must be spanned by two polynomials in $\mathcal{I}$. Note that in this case we get that $P_i = L_i$ is a polynomial defined over V.

Assume then that $\gamma_i = 1$. If P_i is spanned by Q_j and $Q_{j'}$ such that $j, j' \in \mathcal{I}$, then, as before, $\text{Lin}(L_i) \subseteq \text{span}\{a_j b_j, a_{j'} b_{j'}\}$ and hence L_i is a function of the linear forms in V. Thus, the statement holds with $P'_i = L$ and $v_i = c_i = 0$.

The only case left to consider is when $\gamma_i = 1$ and every polynomial Q_j, for $j \in \mathcal{I}$, that satisfies Theorem 5(i) with P_i, does not span with P_i any polynomial in $\{Q_j \mid j \in \mathcal{I}\} \cup \mathcal{L}$. Note that in such a case it must hold that Q_j spans with P_i a polynomial in $\{Q_j \mid j \in [m_1] \setminus \mathcal{I}\} \cup \mathcal{Q}_2$. Observe that since our polynomials are pairwise linearly independent, if two polynomials from $\mathcal{I}$ span the same polynomial with P_i then P_i is in their span and we are done. From

$$|\{Q_j \mid j \in [m_1] \setminus \mathcal{I}\} \cup \mathcal{Q}_2| \leq (m_1 - |\mathcal{I}|) + m_2 \leq 2m_2 < m_1 - m_2 - 2 \leq |\mathcal{I}| - 2 \ ,$$

we see that for P_i to fail to satisfy the claim it must be the case that it satisfies Theorem 5(ii) with at least 2 polynomials whose indices are in $\mathcal{I}$. Let $Q_j, Q_{j'} \in \mathcal{I}$ be two such polynomials. In particular, there are four linear forms c, d, e and f and scalars $\varepsilon_j, \varepsilon_{j'}$, such that

$$P_i - \varepsilon_j Q_j = cd \quad \text{and} \quad P_i - \varepsilon_{j'} Q_{j'} = ef \ . \tag{6}$$

Equivalently,

$$(1 - \varepsilon_j)Q_o = cd + \varepsilon_j a_j b_j - L_i \quad \text{and} \quad (1 - \varepsilon_{j'})Q_o = ef + \varepsilon_{j'} a_{j'} b_{j'} - L_i \ .$$

As $\text{rank}_s(Q_o) \geq 100$

it must hold that $\varepsilon_j = \varepsilon_{j'} = 1$ and hence

$$L_i = cd + a_j b_j \quad \text{and} \quad L_i = ef + a_{j'} b_{j'} \ .$$

It follows that $cd - ef = a_{j'} b_{j'} - a_j b_j$ and therefore $\text{Lin}(cd - ef) \subseteq V$. Theorem 25 implies that without loss of generality $d = \varepsilon_i c + v_i$. We therefore conclude that

$$P_i = Q_o + L_i = Q_o + a_j b_j + c(\varepsilon_i c + v_i)$$

and the statement holds for $P'_i = a_j b_j$ and $c_i = c$. This completes the proof of the Theorem 57. □

Consider the representation guaranteed in Theorem 57 and let

$$\mathcal{S} = \{c_i \mid \text{there is } P_i \in \mathcal{Q} \text{ such that either } P_i = c_i^2 \text{ or, for some } P_i' \text{ defined over } V, \\ P_i = Q_o + P_i' + c_i(\varepsilon_i c_i + v_i)\} \ .$$

Clearly, in order to bound the dimension of $\tilde{\mathcal{Q}}$ it is enough to bound the dimension of $\mathcal{S}$. We do so, by proving that $\mathcal{S}$ satisfies the conditions of Sylvester-Gallai theorem modulo V, and thus have dimension at most $3 + \dim(V) = 7$.

Claim 58. *Let $c_i, c_j \in \mathcal{S}$ be such that $c_i \notin V$ and $c_j \notin \text{span}\{c_i, V\}$. Then, there is $c_k \in \mathcal{S}$ such that $c_k \in \text{span}\{c_i, c_j, V\}$ and $c_k \notin \text{span}\{c_i, V\} \cup \text{span}\{c_j, V\}$.*

Before proving the claim we prove the following simple lemma.

Lemma 59. *Let P_V be a polynomial defined over V and let c_i, c_j as in Theorem 58. If there are linear forms e, f such that*

$$c_j(\varepsilon_j c_j + v_j) + c_i(\varepsilon_i c_i + v_i) + ef = P_V$$

then, without loss of generality, $e \in \text{span}\{c_i, c_j, V\}$ and $e \notin \text{span}\{c_i, V\} \cup \text{span}\{c_j, V\}$.

Proof. First note that $e \notin V$ as otherwise we would have that $c_i \equiv_V c_j$ in contradiction.

By our assumption, $ef = P_V$ modulo c_i, c_j. We can therefore assume without loss of generality that $e \in \text{span}\{c_i, c_j, V\}$. Assume towards a contradiction and without loss of generality that $e = \lambda c_i + v_e$, where $\lambda \neq 0$ and $v_e \in V$. Consider the equation $c_j(\varepsilon_j c_j + v_j) + c_i(\varepsilon_i c_i + v_i) + ef = P_V$ modulo c_i. We have that $c_j(\varepsilon_j c_j + v_j) + v_e f \equiv_{c_i} P_V$ which implies that $\varepsilon_j = 0$. Consequently, we also have that $f = \mu c_j + \eta c_i + v_f$, for some $\mu \neq 0$ and $v_f \in V$. We now observe that the product $c_i c_j$ has a non zero coefficient $\lambda\mu$ in ef and a zero coefficient in $P_V - c_j(\varepsilon_j c_j + v_j) + c_i(\varepsilon_i c_i + v_i)$, in contradiction. □

Proof of Theorem 58. Following the notation of Theorem 57, we either have $Q_i = Q_o + Q_i' + c_i(\varepsilon_i c_i + v_i)$ or $Q_i = c_i^2$. Very similarly to Theorem 52, we consider which case of Theorem 5 Q_i and Q_j satisfy, and what structure they have.

Assume $Q_i = Q_o + Q_i' + c_i(\varepsilon_i c_i + v_i)$ and $Q_j = Q_o + Q_j' + c_j(\varepsilon_j c_j + v_j)$. As argued before, since the rank of Q_o is large they can not satisfy Theorem 5(iii). We consider the remaining cases:

- Q_i, Q_j satisfy Theorem 5(i): there is $Q_k \in \mathcal{Q}$ such that $Q_k \in \text{span}\{Q_i, Q_j\}$. By assumption, for some scalars α, β we have that

  $$Q_k = \alpha(Q_o + Q_i' + c_i(\varepsilon_i c_i + v_i)) + \beta(Q_o + Q_j' + c_j(\varepsilon_j c_j + v_j)) \ . \quad (7)$$

  If Q_k depends only on V then we would get a contradiction to the choice of c_i, c_j. Indeed, in this case we have that

  $$(\alpha + \beta)Q_o = Q_k - \alpha(Q_i' + c_i(\varepsilon_i c_i + v_i)) - \beta(Q_j' + c_j(\varepsilon_j c_j + v_j)) \ .$$

  Rank arguments imply that $\alpha + \beta = 0$ and therefore

  $$\alpha c_i(\varepsilon_i c_i + v_i) + \beta c_j(\varepsilon_j c_j + v_j) = Q_k - \alpha Q_i' - \beta Q_j' \ ,$$

which implies that c_i and c_j are linearly dependent modulo V in contradiction.

If $Q_k = c_k^2$ then by Theorem 59 it holds that c_k satisfies the claim condition.

We therefore assume that Q_k is not a function of V alone and denote $Q_k = Q_o + Q'_k + c_k(\varepsilon_k c_k + v_k)$. Equation 7 implies that

$$(1-\alpha-\beta)Q_o = \alpha Q'_i+\beta Q'_j-Q'_k+\alpha c_i(\varepsilon_i c_i+v_i)+\beta c_j(\varepsilon_j c_j+v_j)-c_k(\varepsilon_k c_k+v_k) \; .$$

As $\alpha Q'_i + \beta Q'_j - Q'_k$ is a polynomial defined over V, its rank is smaller than 4 and thus, combined with the fact that $\text{rank}_s(Q_o) \geq 100$, we get that $(1 - \alpha - \beta) = 0$ and

$$Q'_k - \alpha Q'_i - \beta Q'_j = \alpha c_i(\varepsilon_i c_i + v_i) + \beta c_j(\varepsilon_j c_j + v_j) - c_k(\varepsilon_k c_k + v_k) \; .$$

We now conclude from Theorem 59 that c_k satisfies the claim.

- Q_i, Q_j satisfy Theorem 5(ii): There are linear forms e, f such that for non zero scalars α, β, $\alpha Q_i + \beta Q_j = ef$. In particular,

$$(\alpha + \beta)Q_o = ef - \alpha Q'_i - \beta Q'_j - \alpha c_i(\varepsilon_i c_i + v_i) - \beta c_j(\varepsilon_j c_j + v_j).$$

From rank argument we get that $\alpha + \beta = 0$ and from Theorem 59 we conclude that, without loss of generality, $e = \mu c_i + \eta c_j + v_e$ where $\mu, \eta \neq 0$. We also assume without loss of generality that $Q_i = Q_j + ef$.

By our assumption that $\text{rank}_s(Q_o) \geq 100$ it follows that Q_j is irreducible even after setting $e = 0$. It follows that if a product of irreducible quadratics satisfy

$$\prod_k A_k \in \sqrt{\langle Q_i, Q_j \rangle} = \sqrt{\langle ef, Q_j \rangle}$$

then, after setting $e = 0$, some A_k is divisible by $Q_j|_{e=0}$. Thus, there is a multiplicand that is equal to $\gamma Q_j + ed$ for some linear form d and scalar γ. In particular, there must be a polynomial $Q_k \in \tilde{\mathcal{Q}} \setminus \{Q_1, Q_2\}$, such that $Q_k = \gamma Q_j + ed$. If $\gamma = 0$ then it must hold that $Q_k = a_k^2 = ed$ and thus $a_k \sim e$, and the statment holds. If $\gamma = 1$ then we can assume without loss of generality that $Q_k = Q_j + ed$. Thus,

$$Q+Q'_k+c_k(\varepsilon_k c_k+v_k) = Q_k = Q_j+ed = Q_o+Q'_j+c_j(\varepsilon_j c_j+v_j)+(\mu c_i+\eta c_j+v_e)d \; .$$

Setting $c_j = 0$ we get that

$$Q'_k + c_k(\varepsilon_k c_k + v_k) \equiv_{c_j} Q'_j + (\mu c_i + v_e)d \; . \tag{8}$$

Note that it cannot be the case that $d \equiv_{c_j} 0$. Indeed, if $d = 0$ then we get that Q_j and Q_k are linearly dependent in contradiction. If $d \sim c_j$ then (8) implies that $c_k \in \text{span}\{c_j, V\}$. From the equality $Q_k = Q_j + ed$ and the fact that e depends non trivially on c_i, it now follows that $c_i \in \text{span}\{c_j, V\}$ in contradiction to the choice of c_i and c_j. As $d \not\equiv_{c_j} 0$, we deduce from (8) that, modulo c_j, $c_k \in \text{span}\{c_i, V\}$. We next show that if c_k depends only on c_i and V then we reach a contradiction and this will conclude the

proof. So assume towards a contradiction that $c_k = \lambda c_i + v'_k$, for a scalar λ and $v'_k \in V$. Since

$$Q_j + ed = Q_k = Q_o + Q'_k + c_k(\varepsilon_k c_k + v_k) = Q_o + Q'_k + (\lambda c_i + v'_k)\,(\varepsilon_k(\lambda c_i + v'_k) + v_k)$$

and

$$Q_j + ef = Q_i = Q_o + Q'_i + c_i(\varepsilon_i c_i + v_i)$$

we get by subtracting Q_i from Q_k that

$$e(d-f) = Q_k - Q_i = Q'_k - Q'_i + (\lambda c_i + v'_k)\,(\varepsilon_k(\lambda c_i + v'_k) + v_k) - c_i(\varepsilon_i c_i + v_i)$$

and clearly neither side of the equation is zero since $Q_i \neq Q_k$. This implies that $e \in \text{span}\{c_i, V\}$. This however contradicts the fact that $e = \mu c_i + \eta c_j + v_e$ where $\mu, \eta \neq 0$.

Now let us consider the case where without loss of generality, $Q_i = Q_o + Q'_i + c_i(\varepsilon_i c_i + v_i)$ and $Q_j = c_j^2$. In this case the polynomials satisfy Theorem 5(ii) as $0 \cdot Q_i + Q_j = c_j^2$. Similarly to the previous argument, it holds that there is Q_k such that $Q_k = \gamma Q_i + c_j e$. If $\gamma = 0$ it holds that Q_k is reducible, and therefore a square of a linear form, in contradiction to pairwise linear independence. Thus $\gamma \neq 0$. If Q_k is defined only on the linear functions in V then it is of rank smaller then $\dim(V) \leq 4$, which will result in a contradiction to the rank assumption on Q_o. Thus $Q_k = Q_o + Q'_k + c_k(\varepsilon_k c_k + v_k)$ and $\gamma = 1$. Therefore, we have

$$Q_o + Q'_k + c_k(\varepsilon_k c_k + v_k) = Q_k = Q_i + c_j e \;= Q_o + Q'_i + c_i(\varepsilon_i c_i + v_i) + c_j e.$$

Hence,

$$Q'_k - Q'_i - c_i(\varepsilon_i c_i + v_i) - c_j e = -c_k(\varepsilon_k c_k + v_k).$$

Looking at this equation modulo c_j implies that $c_k \in \text{span}\{V, c_i, c_j\}$. and $c_k \notin \text{span}\{V, c_j\}$, or we will get a contradiction to the fact that $c_i \notin \text{span}\{c_j, V\}$. Similarly it holds that $c_k \notin \text{span}\{V, c_i\}$, as we wanted to show.

The last structure we have to consider is the case where $Q_i = c_i^2, Q_j = c_j^2$. In this case, the ideal $\sqrt{\langle c_i^2, c_j^2\rangle} = \langle c_i, c_j\rangle$ is prime and therefore there is $Q_k \in \langle c_i, c_j\rangle$ this means that $\text{rank}_s(Q_k) \leq 2$. If $\text{rank}_s(Q_k) = 1$ then $Q_k = c_k^2$ and the statement holds. $\text{rank}_s(Q_k) = 2$ then Q_k is defined on the linear function of V, which implies $c_i, c_j \in V$ in contradiction to our assumptions. □

We are now ready to prove Theorem 46.

Proof of Theorem 46. Theorem 58 implies that if we project the linear forms in $\mathcal{S}$ to $V^\perp$ then, after removing linearly dependent forms, they satisfy the conditions of the Sylvester-Gallai theorem. As $\dim(V) \leq 4$ we obtain that $\dim(\text{span}\{\mathcal{S} \cup V\}) \leq 7$. By Theorem 57 every polynomial $P \in \mathcal{Q}$ is a linear combination of Q_o and a polynomial defined over $\text{span}\{\mathcal{S} \cup V\}$ which, by the argument above, implies that $\dim(\text{span}\{\mathcal{Q}\}) \leq 8$. □

This completes the proof of Theorem 44 when Q_o has high rank. We next handle the case where Q_o is of low rank.

4.2 Q_o is of Low Rank

In this section we prove the following claim.

Claim 60. *Let $\tilde{\mathcal{Q}}$ be a (Q_o, m_1, m_2)-set such that $2 \leq \text{rank}_s(Q_o) < 100$. Then, $\dim(\text{span}\{\tilde{\mathcal{Q}}\}) = O(1)$.*

Before we start with the proof of the main claim, let us prove a similar claim but for a more specific structure of polynomials. We will later see that, essentially, this structure holds when $2 \leq \text{rank}_s(Q_o) < 100$.

Claim 61. *Let $\tilde{\mathcal{Q}}$ be a set of quadratics polynomials that satisfy the conditions in the statement of Theorem 4. Assume farther that there is a linear space of linear forms, V such that $\dim(V) = \Delta$ and for each polynomial $Q_i \in \tilde{\mathcal{Q}}$ one of the following holds: either $Q_i \in \langle V \rangle$ or there is a linear form a_i such that $Lin(Q_i) \subseteq \text{span}\{V, a_i\}$. Then $\dim(\tilde{\mathcal{Q}}) \leq 8\Delta^2$.*

Proof. Note that by the conditions in the statement of Theorem 4, no two polynomials in $\tilde{\mathcal{Q}}$ share a common factor.

Let $\boldsymbol{\alpha} \in \mathbb{C}^{\Delta}$ be such that if two polynomials in $T_{\boldsymbol{\alpha},V}(\tilde{\mathcal{Q}})$ (recall Theorem 27) share a common factor then it is a polynomial in z. Note that by Theorem 29 such $\boldsymbol{\alpha}$ exists. Thus, each $P \in \tilde{\mathcal{Q}}$, satisfies that either $T_{\boldsymbol{\alpha},V}(P) = \alpha_P z^2$ or $\text{Lin}(T_{\boldsymbol{\alpha},V}(P)) \subseteq \text{span}\{z, a_P\}$ for some linear form a_P independent of z. It follows that every polynomial in $T_{\boldsymbol{\alpha},V}(\tilde{\mathcal{Q}})$ is reducible. We next show that $\mathcal{S} = \{a_P \mid P \in \tilde{\mathcal{Q}}\}$ satisfies the conditions of Sylvester-Gallai theorem modulo z.

Let $a_1, a_2 \in \mathcal{S}$ such that $a_2 \notin \text{span}\{z, a_1\}$. Consider Q_1 such that $\text{Lin}(T_{\boldsymbol{\alpha},V}(Q_1)) \subseteq \text{span}\{z, a_1\}$ yet $\text{Lin}(T_{\boldsymbol{\alpha},V}(Q_1)) \not\subseteq \text{span}\{z\}$. Similarly, let Q_2 be such that $\text{Lin}(T_{\boldsymbol{\alpha},V}(Q_2)) \subseteq \text{span}\{z, a_2\}$ and $\text{Lin}(T_{\boldsymbol{\alpha},V}(Q_2)) \not\subseteq \text{span}\{z\}$. Then there is a factor of $T_{\boldsymbol{\alpha},V}(Q_1)$ of the form $\gamma_1 z + \delta_1 a_1$ where $\delta_1 \neq 0$. Similarly there is a factor of $T_{\boldsymbol{\alpha},V}(Q_2)$ of the form $\gamma_2 z + \delta_2 a_2$ where $\delta_2 \neq 0$.

This implies that $\sqrt{\langle T_{\boldsymbol{\alpha},V}(Q_1), T_{\boldsymbol{\alpha},V}(Q_2)\rangle} \subseteq \langle \gamma_1 z + \delta_1 a_1, \gamma_2 z + \delta_2 a_2 \rangle$. Indeed, it is clear that for $i \in \{1, 2\}$, $T_{\boldsymbol{\alpha},V}(Q_i) \in \langle \gamma_i z + \delta_i a_i \rangle$. Hence, $\sqrt{\langle T_{\boldsymbol{\alpha},V}(Q_1), T_{\boldsymbol{\alpha},V}(Q_2)\rangle} \subseteq \sqrt{\langle \gamma_1 z + \delta_1 a_1, \gamma_2 z + \delta_2 a_2 \rangle} = \langle \gamma_1 z + \delta_1 a_1, \gamma_2 z + \delta_2 a_2 \rangle$, where the equality holds since $\langle \gamma_1 z + \delta_1 a_1, \gamma_2 z + \delta_2 a_2 \rangle$ is a prime ideal.

We know that, there are $Q_3, Q_4, Q_5, Q_6 \in \mathcal{Q}$ such that

$$Q_3 \cdot Q_4 \cdot Q_5 \cdot Q_6 \in \sqrt{\langle Q_1, Q_2 \rangle}.$$

As $T_{\boldsymbol{\alpha},V}$ is a ring homomorphism it follows that,

$$T_{\boldsymbol{\alpha},V}(Q_3) \cdot T_{\boldsymbol{\alpha},V}(Q_4) \cdot T_{\boldsymbol{\alpha},V}(Q_5) \cdot T_{\boldsymbol{\alpha},V}(Q_6) \in \sqrt{\langle T_{\boldsymbol{\alpha},V}(Q_1), T_{\boldsymbol{\alpha},V}(Q_2)\rangle} \subseteq \langle \gamma_1 z + \delta_1 a_1, \gamma_2 z + \delta_2 a_2 \rangle .$$

Since $\langle \gamma_1 z + \delta_1 a_1, \gamma_2 z + \delta_2 a_2 \rangle$ is prime it follows that, without loss of generality, $T_{\boldsymbol{\alpha},V}(Q_3) \in \langle \gamma_1 z + \delta_1 a_1, \gamma_2 z + \delta_2 a_2 \rangle$. It cannot be the case that $T_{\boldsymbol{\alpha},V}(Q_3) \in \langle \gamma_i z + \delta_i a_i \rangle$ for any $i \in \{1, 2\}$, because otherwise this will imply that $T_{\boldsymbol{\alpha},V}(Q_3)$ and $T_{\boldsymbol{\alpha},V}(Q_i)$ share a common factor that is not a polynomial in z, in contradiction to our choice of $T_{\boldsymbol{\alpha},V}$. This means that there is a factor of $T_{\boldsymbol{\alpha},V}(Q_3)$ that is in $\text{span}\{a_1, a_2, z\} \setminus (\text{span}\{a_1, z\} \cup \text{span}\{a_2, z\})$. Consequently, $a_3 \in \text{span}\{a_1, a_2, z\} \setminus (\text{span}\{a_1, z\} \cup \text{span}\{a_2, z\})$ as we wanted to prove. This shows that $\mathcal{S}$ satisfies the conditions of Sylvester-Gallai theorem, and therefore $\dim(\mathcal{S}) \leq 3$. Repeating the analysis above for linearly independent $\boldsymbol{\alpha}_1, \ldots, \boldsymbol{\alpha}_{\Delta}$, we can

use Theorem 32 and obtain that $\dim(\mathrm{Lin}(\tilde{\mathcal{Q}})) \leq (3+1)\Delta$, and thus $\dim(\tilde{\mathcal{Q}}) \leq \binom{4\Delta}{2} + \Delta \leq 8\Delta^2$. □

Back to the proof of Theorem 60. As before we first prove the claim for the case $m_2 = 0$ and then we prove the general case.

4.2.1 The case $m_2 = 0$

Similarly to the high rank case, in this subsection we prove the following claim.

Claim 62. *Let $\tilde{\mathcal{Q}} = \mathcal{Q} \cup \mathcal{L}$ be $a(Q_o, m_1, 0)$-set such that $2 \leq \mathrm{rank}_s(Q_o) < 100$, then $\dim(\mathrm{span}\{a_1, \ldots, a_{m_1}, b_1, \ldots, b_{m_1}, \ell_1, \ldots, \ell_r\}) = O(1)$.*

The proof is similar in structure to the proof of Theorem 47. As before, we consider a polynomial $\ell_i^2 \in \mathcal{L}$ as $0 \cdot Q_o + \ell_i \ell_i$. We start by proving an analog of Theorem 49. The claims are similar but the proofs are slightly different as we cannot rely on Q_o having high rank.

Claim 63. *Let $\tilde{\mathcal{Q}}$ satisfy the assumptions of Theorem 62. Let $i \in [m_1]$ be such that $dim(a_i, b_i) = 2$ and $\mathrm{span}\{a_i, b_i\} \cap Lin(Q_o) = \{\mathbf{0}\}$. Then, for every $j \in [m_1]$ the following holds:*

1. *Q_i and Q_j do not satisfy Theorem 5(iii).*
2. *If Q_i and Q_j satisfy Theorem 5(i) then there exists $\alpha, \beta \in \mathbb{C} \setminus \{0\}$ such that for some $k \in [m_1] \setminus \{i, j\}$*

$$\alpha a_i b_i + \beta a_j b_j = a_k b_k \,. \tag{9}$$

3. *If Q_j is irreducible and Q_i and Q_j satisfy Theorem 5(ii) then there exist two linear forms, c and d such that*

$$a_i b_i - a_j b_j = cd \,. \tag{10}$$

Proof. Assume Q_i and Q_j satisfy Theorem 5(i), i.e., there are $\alpha, \beta \in \mathbb{C}$ and $k \in [m_1] \setminus \{i, j\}$ such that

$$\alpha(Q_o + a_i b_i) + \beta(Q_o + a_j b_j) = \alpha Q_i + \beta Q_j = Q_k = \alpha_k Q + a_k b_k$$

This implies that $\alpha a_i b_i + \beta a_j b_j - a_k b_k = (\alpha_k - (\alpha + \beta))Q_o$. We next show that it must be the case that $\alpha_k - (\alpha + \beta) = 0$.

Indeed, if $\alpha_k - (\alpha+\beta) \neq 0$ we get that $\beta a_j b_j - a_k b_k = (\alpha_k - (\alpha+\beta))Q_o - \alpha a_i b_i$. However, as we assumed $\mathrm{span}\{a_i, b_i\} \cap \mathrm{Lin}(Q_o) = \{\mathbf{0}\}$, we get by Theorem 23 that

$$\mathrm{rank}_s(\alpha_k - (\alpha + \beta))Q_o - \alpha a_i b_i) = \mathrm{rank}_s(Q_o) + 1 > 2 \geq \mathrm{rank}_s(\beta a_j b_j - a_k b_k)$$

in contradiction. We thus have that $\alpha_k - (\alpha + \beta) = 0$ and hence

$$\alpha a_i b_i + \beta a_j b_j = a_k b_k \tag{11}$$

and Equation 9 is satisfied. Observe that since our polynomials are pairwise independent $\alpha, \beta \neq 0$.

A similar argument to the one showing $\alpha_k - (\alpha + \beta) = 0$ also implies that Q_i and Q_j do not satisfy Theorem 5(iii). If this was not the case then we would have that $\text{rank}_s(Q_o + a_i b_i) = 2$ which would again contradict Theorem 23.

If Q_j is irreducible, the only case left is when $Q_o + a_i b_i, Q_o + a_j b_j$ satisfy Theorem 5(ii). In this case there are $\alpha, \beta \in \mathbb{C}$ and two linear forms c and d such that $\alpha(Q_o + a_i b_i) + \beta(Q_o + a_j b_j) = cd$, and again, by the same argument we get that $\beta = -\alpha$ and so (after rescaling c)

$$a_i b_i - a_j b_j = cd \ .$$

This completes the proof of Theorem 63. □

For each $i \in [m_1]$ let $V_i := \text{span}\{a_i, b_i\}$. The next claim is analogous to Theorem 50.

Claim 64. *Let $\tilde{\mathcal{Q}}$ satisfy the assumption in Theorem 62. If for some $i \in [m_1]$ it holds that $\dim(V_i) = 2$ and $Lin(Q_o) \cap V_i = \{\mathbf{0}\}$ then for every $j \in [m_1]$ it is the case that $\dim(V_j \cap V_i) \geq 1$. In particular, if $dim(V_j) = 1$ then $V_j \subsetneq V_i$.*

Proof. The proof of this claim follows immediately from Theorem 63 and Theorem 24. □

the next claim is an analogous to Theorem 51.

Claim 65. *Under the assumptions of Theorem 62 there exists a subspace V of linear forms such that $\dim(V) \leq 2 \cdot 100 + 3$ and for every $i \in [m_1]$ there exists $v_i \in V$ and a constant $\varepsilon_i \in \mathbb{C}$ such that $b_i = \varepsilon_i a_i + v_i$ (or $a_i = \varepsilon_i b_i + v_i$).*

Proof. Let $\mathcal{I} = \{i \in [m_1] \mid \dim(V_i) = 2 \text{ and } \text{Lin}(Q_o) \cap V_i = \{\mathbf{0}\}\}$. If $\dim(\bigcup_{i \in \mathcal{I}} V_i) \leq 3$ then we set $V = \text{span}\{\text{Lin}(Q_o) \cup (\bigcup_{i \in \mathcal{I}} V_i)\}$. Clearly $\dim(V) \leq 2 \cdot \text{rank}_s(Q) + 3 \leq 2 \cdot 100 + 3$. Theorem 64 implies that V has the required properties.

If $\dim(\bigcup_{i \in \mathcal{I}} V_i) > 3$ then from Theorem 64 and Theorem 26 it follows that $\dim(\bigcap_{i \in \mathcal{I}} V_i) = 1$. Let w be such that $\text{span}\{w\} = \bigcap_{i \in \mathcal{I}} V_i$ and set $V = \text{span}\{\text{Lin}(Q_o), w\}$. In this case too it is easy to see that V has the required properties.

□

From now on we assume, without loss of generality that for every $i \in [m_1]$, $b_i = \varepsilon_i a_i + v_i$. This structure also holds for the polynomials in $\mathcal{L}$.

Proof of Theorem 62. Theorem 65 implies that there is a linear space of linear forms, V, with $\dim(V) \leq 2 \cdot 100 + 3$, with the property that for every $Q_i \in \tilde{\mathcal{Q}}$ there is a linear form a_i such that $\text{Lin}(Q_i) \subseteq \text{span}\{V, a_i\}$.

Thus $\tilde{\mathcal{Q}}$ satisfies the conditions of Theorem 61, and $\dim(\tilde{\mathcal{Q}}) = O(1)$, as we wanted to show. □

We next consider the case $m_2 \neq 0$.

4.2.2 The case $m_2 \neq 0$

In this subsection we prove Theorem 60, we can assume without loss of generality that $m_2 \neq 0$, as the case that $m_2 = 0$ was proved in the previous subsection. To handle this case we prove the existence of a subspace V of linear forms, of dimension $O(1)$, such that every polynomial in $\tilde{\mathcal{Q}}$ is in $\langle V \rangle$, and then, like we did before, we bound the dimension of $\tilde{\mathcal{Q}}$. The first step is proving an analog of Theorem 54.

Claim 66. *Let $\tilde{\mathcal{Q}}$ be a (Q_o, m_1, m_2)-set such that $\mathrm{rank}_s(Q_o) < 100$. Then for every $i \in [m_2]$ there exists $\gamma_i \in \mathbb{C}$ such that $\mathrm{rank}_s(P_i - \gamma_i Q_o) = 2$.*

Proof. Consider $i \in [m_2]$. If P_i satisfies Theorem 5(iii) with any $Q_j \in \mathcal{Q}_1$, then the claim holds with $\gamma_i = 0$. If P_i satisfies Theorem 5(ii) with any $Q_j \in \mathcal{Q}$ then there exist linear forms c and d and non zero $\alpha, \beta \in \mathbb{C}$, such that $\alpha P_i + \beta Q_j = cd$. Therefore, $P_i = \frac{1}{\alpha}(cd - \beta(Q + a_j b_j))$ and the statement holds with $\gamma_i = -\frac{\beta}{\alpha}$. Observe that the rank of $cd - \beta a_j b_j$ cannot be 1 by Theorem 43.

Thus, the only case left to consider is when P_i satisfies Theorem 5(i) with all the Q_j's in $\mathcal{Q}_1$. We next show that in this case there must exist $j \neq j' \in [m_1]$ such that $Q_{j'} \in \mathrm{span}\{Q_j, P_i\}$. Indeed, since $m_1 > 5m_2 + 2$ there must be $j, j' \in [m_1]$ and $i' \in [m_2]$ such that $P_{i'} \in \mathrm{span}\{Q_{j'}, P_i\}$ and $P_{i'} \in \mathrm{span}\{Q_j, P_i\}$. As we saw before this implies that $P_i \in \mathrm{span}\{Q_j, Q_{j'}\}$, which is what we wanted to show.

Let $j \neq j' \in [m_1]$ be as above and let $\alpha, \beta \in \mathbb{C}$ be such that $P_i = \alpha Q_j + \beta Q_{j'}$. It follows that

$$P_i = (\alpha + \beta)Q_o + \alpha a_j b_j + \beta a_{j'} b_{j'} \ .$$

Let $\gamma_i = \alpha + \beta$. Property 4 in Theorem 43 implies that $\mathrm{rank}_s(\alpha a_j b_j + \beta a_{j'} b_{j'}) = 2$ and the claim follows. □

As before, whenever $\gamma_i \neq 0$ let us replace P_i with $\frac{1}{\gamma_i} P_i$. Thus, from now on we shall assume $\gamma_i \in \{0, 1\}$. We next prove an analog of Theorem 55.

Claim 67. *Let $\tilde{\mathcal{Q}}$ be a (Q_o, m_1, m_2)-set such that $\mathrm{rank}_s(Q_o) < 100$. Then there is a subspace V of linear forms such that $\dim(V) \leq 2 \cdot 100 + 4$, $Lin(Q_o) \subseteq V$ and for at least $m_1 - 2m_2$ of the indices $j \in [m_1]$ it holds that $a_j, b_j \in V$.*

Proof. Let $P = P_1$. Theorem 66 implies that $P = \gamma Q_o + L$, for some L of rank 2. Set $V = \mathrm{span}\{\mathrm{Lin}(Q_o) \cup \mathrm{Lin}(L)\}$. Clearly $\dim(V) \leq 2 \cdot 100 + 4$.

Let $j \in [m_1]$. If P and Q_j satisfy Theorem 5(iii), then there are two linear forms c and d such that $Q_j, P \in \sqrt{\langle c, d \rangle}$, this implies that $\mathrm{span}\{c, d\} \subset \mathrm{Lin}(P) \subseteq V$.

If $Q_o = Q_j - a_j b_j$ is not zero modulo c, d, then we obtain that $Q_o \equiv_{c,d} -a_j b_j$. Thus, there are linear forms $v_1, v_2 \in \mathrm{Lin}(Q_o)$ such that $a_j \equiv_{c,d} v_1$ and $b_j \equiv_{c,d} v_2$. In particular, as $\mathrm{Lin}(Q_o) \cup \{c, d\} \subset V$ it follows that $a_j, b_j \in V$. If Q_o is zero modulo c and d, then Q_j, Q_o satisfy Theorem 5(iii) and from property 5 of Theorem 43 we know that there are at most m_2 such Q_j's. Furthermore, as $c, d \in \mathrm{Lin}(Q_o) \subset V$ we obtain that $Q_j \in \langle V \rangle$. Denote by $\mathcal{K}$ the set of all Q_j that satisfy Theorem 5(iii) with Q_o. As we mentioned, $|\mathcal{K}| \leq m_2$.

If P and Q_j satisfy Theorem 5(ii) then there are two linear forms c and d, and non zero $\alpha, \beta \in \mathbb{C}$, such that $\alpha P + \beta Q_j = cd$. Hence,

$$\beta Q_o + \alpha P = -\beta a_j b_j + cd \ .$$

As $\beta Q_o + \alpha P$ is a non trivial linear combination of Q_o and P, we get from property 4 of Theorem 43 that $2 \leq \text{rank}_s((\alpha\gamma + \beta)Q_o + \alpha L)$. It follows that

$$\text{rank}_s(-\beta a_j b_j + cd) = \text{rank}_s((\alpha\gamma + \beta)Q_o + \alpha L) = 2$$

and therefore by Theorem 21,

$$\{a_j, b_j, c, d\} \subset \text{Lin}(-\beta a_j b_j + cd) = \text{Lin}((\alpha\gamma + \beta)Q_o + \alpha L) \subseteq V \ ,$$

and again $a_j, b_j \in V$.

The last case to consider is when P and Q_j satisfy Theorem 5(i). If they span a polynomial $Q_{j'} \in \mathcal{Q}_1 \cup \mathcal{L}$, then $P = \alpha Q_j + \beta Q_{j'}$ and as in the previous case we get that $a_j, b_j \in V$.

Let $\mathcal{J}$ be the set of all indices $j \in [m_1]$ such that P and Q_j span a polynomial in $\mathcal{Q}_2$ but no polynomial in $\mathcal{Q}_1 \cup \mathcal{L}$. So far we proved that for every $j \in [m_1] \setminus (\mathcal{J} \cup \mathcal{K})$ we have that $a_j, b_j \in V$. We next show that $|\mathcal{J}| \leq m_2$ which concludes the proof.

Indeed, if this was not the case then by the pigeonhole principle there would exist a polynomial $P_i \in \mathcal{Q}_2$ and two polynomials $Q_j, Q_{j'} \in \mathcal{Q}_1$ such that $P_i \in \text{span}\{Q_j, P\}$ and $P_i \in \text{span}\{Q_{j'}, P\}$. By pairwise independence this implies that $Q_{j'}$ is in the linear span of P and Q_j which contradicts the definition of $\mathcal{J}$. □

Our next claim gives more information about the way the polynomials in $\tilde{\mathcal{Q}}$ relate to the subspace V found in Theorem 67.

Claim 68. *Let $\tilde{\mathcal{Q}}$ and V be as in Theorem 67. Then, every polynomial P in $\tilde{\mathcal{Q}}$ satisfies (at least) one of the following cases:*

1. *Lin$(P) \subseteq V$ or*
2. *$P \in \langle V \rangle$ or*
3. *$P = P' + c(c+v)$ where P' is a quadratic polynomial such that Lin$(P') \subseteq V$, $v \in V$ and c is a linear form.*

Proof. Let $\mathcal{I} = \{j \in [m_1] \mid a_j, b_j \in V\}$. Theorem 67 implies that $|\mathcal{I}| \geq m_1 - 2m_2$. Furthermore, by the construction of V we know that $\text{Lin}(Q_o) \subseteq V$. Observe that this implies that for every $j \in \mathcal{I}$, $\text{Lin}(Q_j) \subseteq V$.

Note that every polynomial in $\mathcal{L}$ satisfies the third item of the claim. Let P be any polynomial in $\mathcal{Q}_2 \cup \{Q_j \mid j \in [m_1] \setminus \mathcal{I}\}$. We study which case of Theorem 5 P satisfies with polynomials whose indices belong to $\mathcal{I}$.

If P_i satisfies Theorem 5(iii) with any polynomial Q_j, for $j \in \mathcal{I}$, then, as $\text{Lin}(Q_j) \subseteq V$, it follows that $P \in \langle V \rangle$.

If P is spanned by two polynomials $Q_j, Q_{j'}$ such that $j, j' \in \mathcal{I}$, then clearly $\text{Lin}(P) \subseteq V$. Similarly, if P is spanned by a polynomial $Q_j, Q_{j'}$ such that $j \in \mathcal{I}$ and $Q_{j'} \in \mathcal{L}$ then $P = \alpha Q_j + \beta a_{j'}^2$, and hence it also satisfies the claim.

Hence, for P to fail to satisfy the claim, it must be the case that every polynomial Q_j, for $j \in \mathcal{I}$, that satisfies Theorem 5(i) with P, does not span with P any polynomial in $\{Q_j \mid j \in \mathcal{I}\} \cup \mathcal{L}$. Thus, it must span with P a polynomial in $\{Q_j \mid j \in [m_1] \setminus \mathcal{I}\} \cup \mathcal{Q}_2$. As before, observe that by pairwise

linear independent, if two polynomials from $\mathcal{I}$ span the same polynomial with P, then P is in their span and we are done. Thus, since

$$|\{Q_j \mid j \in [m_1] \setminus \mathcal{I}\} \cup \mathcal{Q}_2| \leq (m_1 - |\mathcal{I}|) + m_2 \leq 3m_2 < m_1 - 2m_2 - 2 \leq |\mathcal{I}| - 2 \,,$$

for P to fail to satisfy the claim it must be the case that it satisfies Theorem 5(ii) with at least 2 polynomials whose indices are in $\mathcal{I}$.

Let $Q_j, Q_{j'}$ be two such polynomials. There are four linear forms, c, d, e and f and scalars $\varepsilon_j, \varepsilon_{j'}$ such that

$$P + \varepsilon_j Q_j = cd \qquad \text{and} \qquad P + \varepsilon_{j'} Q_{j'} = ef \,.$$

Therefore

$$\varepsilon_j Q_j - \varepsilon_{j'} Q_{j'} = cd - ef \,. \tag{12}$$

In particular, $\mathrm{Lin}(cd - ef) \subseteq V$. Theorem 25 and Equation (12) imply that, without loss of generality, $d = \varepsilon c + v$ for some $v \in V$ and $\varepsilon \in \mathbb{C}$. Thus, $P = cd - \varepsilon_j Q_j = c(\varepsilon c + v) - \varepsilon_j Q_j$ and no matter whether $\varepsilon = 0$ or not. P satisfies the claim. Indeed, if $\varepsilon = 0$ then $P \in \langle V \rangle$ and we are done. Otherwise, we can normalize c, v to assume that $\varepsilon = 1$ and get that $\mathrm{Lin}(P - c^2) \in V$ as claimed.

□

We can now complete the proof of Theorem 60.

Proof of Theorem 60. Theorem 68 implies that there is a linear space of linear forms, V, such that $\dim(V) \leq 2 \cdot 100 + 4$ and every polynomial $Q_i \in \tilde{\mathcal{Q}}$ satisfies the following. Either $Q_i \in \langle V \rangle$ or, there is a linear form a_i such that $\mathrm{Lin}(Q_i) \subseteq \mathrm{span}\{V, a_i\}$. (It might be that $\mathrm{Lin}(Q_i) \subseteq V$ or that $\mathrm{Lin}(Q_i) \subseteq \mathrm{span}\{a_i\}$). Thus $\tilde{\mathcal{Q}}$ satisfies the conditions of Theorem 61, and $\dim(\tilde{\mathcal{Q}}) = O(1)$, as we wanted to show.

□

Theorem 46 together with Theorem 60 completes the proof of Theorem 44.
□

Journal Section3: Structure theorem for quadratics

Shir Peleg and Amir Shpilka, Forum of Mathematics, Sigma10:e112 (2022), pp11–16, DOI10.1017/fms.2022.100. CC BY4.0. The numbering below follows the journal version. Editorial reference-number concordance: journal reference [12] is Gupta (2014), listed as CCC reference [17] in this material. Journal Theorem2.11 is the resultant criterion stated as CCC Theorem17 in the preceding context. The journal continuation retains its original numbering.

Theorem (Author Theorem3.1)**.** *Let $\{Q_k\}_{k\in\mathcal{K}}, A, B$ be homogeneous polynomials of degree 2 such that $\prod_{k\in\mathcal{K}} Q_k \in \sqrt{\langle A, B\rangle}$. Then one of the following cases hold:*

(prime-case) *There is $k \in \mathcal{K}$ such that Q_k is in the linear span of A, B.*

(product-case) *There exists a nontrivial linear combination of the form* $\alpha A + \beta B = c \cdot d$*, where* c *and* d *are linear forms.*

(linear-case) *There exist two linear forms* c *and* d *such that when setting* $c = d = 0$ *we get that* A, B *and one of* $\{Q_k\}_{k\in\mathcal{K}}$ *vanish.*

The following claim of [12] shows that we can assume $|\mathcal{K}| = 4$ in the statement of Theorem3.1.

Claim (Author Claim3.2, Claim11 in journal reference12)**.** *Let* $P_1, \ldots, P_d, Q_1, \ldots, Q_k \in \mathbb{C}[x_1, \ldots, x_n]$ *be homogeneous and the degree of each* P_i *is at most* r*. Then,*

$$\prod_{i=1}^{k} Q_i \in \sqrt{\langle P_1, \ldots, P_d\rangle} \;\Rightarrow\; \exists\{i_1, \ldots, i_{r^d}\} \subset [k] \text{ such that } \prod_{j=1}^{r^d} Q_{i_j} \in \sqrt{\langle P_1, \ldots, P_d\rangle}.$$

Thus, for $r = d = 2$ it follow that there are at most four polynomials among the Q_is whose product is in $\sqrt{\langle A, B\rangle}$. Before proving Theorem3.1, we explain the intuition behind the different cases in the theorem. Clearly, if one of $Q_1, \ldots, Q_4$ is a linear combination of A, B, then it is in their radical (and in fact, in their linear span). If A and B span a product of the form ab, then, say, $(A + ac)(A + bd)$ is in their radical. Indeed, $\sqrt{\langle A, B\rangle} = \sqrt{\langle A, ab\rangle}$. This case is clearly different than the linear span case. Finally, we note that, if $A = ac + bd$ and $B = ae + bf$, then the product $a \cdot b \cdot (cf - de)$ is in $\sqrt{\langle A, B\rangle}$.[5] This case is different than the other two cases as A and B do not span any linear form (or any reducible quadratic) non trivially. Thus, all the three cases are distinct and can happen. What Theorem3.1 shows is that, essentially, these are the only possible cases.

Author proof of Theorem3.1. Following Theorem3.2, we shall assume in the proof that $|\mathcal{K}| = 4$. By applying a suitable linear transformation, we can assume that for some $r \geq 1$

$$A = \sum_{i=1}^{r} x_i^2.$$

We can also assume without loss of generality that x_1^2 appears only in A as we can replace B with any polynomial of the form $B' = B - \alpha A$ without affecting the result as $\langle A, B\rangle = \langle A, B'\rangle$. Furthermore, all cases in the theorem remain the same if we replace B with B' and vice versa. In a similar fashion, we can replace Q_1 with $Q_1' = Q_1 - \alpha A$ to get rid of the term x_1^2 in Q_1. We can do the same for the other Q_is. Thus, without loss of generality, the situation is

$$\begin{aligned} A &= x_1^2 - A' \\ B &= x_1 \cdot b - B' \\ Q_i &= x_1 \cdot b_i - Q_i' \quad \text{for } i \in \{1, 2, 3, 4\}, \end{aligned} \tag{3.1}$$

where A', b, B', Q_i', b_i are homogeneous polynomials that do not depend on x_1, the polynomials A', B', Q' have degree ≤ 2 and b, b_i are linear forms. The analysis shall deal with two cases according to whether B depends on x_1 or not, as we only consider the resultant of A and B with respect to x_1 when it appears in both polynomials.

[5] If we insist on having all factors of degree2, then the same argument shows that the product $(a^2 + A) \cdot (b^2 + B) \cdot (cf - de)$ is in $\sqrt{\langle A, B\rangle}$.

Case $b \not\equiv 0$ Consider the resultant of A and B with respect to x_1. It is easy to see that

$$\mathrm{Res}_{x_1}(A,B) = B'^2 - b^2 \cdot A'.$$

We first prove that, if the resultant is irreducible, then Case(prime-case) of Theorem3.1 holds. For this, we shall need the following claim.

Claim (Author Claim3.3)**.** *Whenever* $\mathrm{Res}_{x_1}(A,B) = 0$*, it holds that* $\prod_{i=1}^{4}(B' \cdot b_i - b \cdot Q_i') = 0$.

Proof. Let $\boldsymbol{\alpha} \in \mathbb{C}^{n-1}$ be such that $\mathrm{Res}_{x_1}(A,B)(\boldsymbol{\alpha}) = 0$ then either $b(\boldsymbol{\alpha}) = 0$, which also implies $B'(\boldsymbol{\alpha}) = 0$ and in this case the claim clearly holds, or $b(\boldsymbol{\alpha}) \neq 0$. Consider the case $b(\boldsymbol{\alpha}) \neq 0$, and set $x_1 = B'(\boldsymbol{\alpha})/b(\boldsymbol{\alpha})$ (we are free to select a value for x_1 as $\mathrm{Res}_{x_1}(A,B)$ does not involve x_1). Notice that for this substitution we have that $B(\boldsymbol{\alpha}) = 0$ and that

$$A|_{x_1 = B'(\boldsymbol{\alpha})/b(\boldsymbol{\alpha})} = (B'(\boldsymbol{\alpha})/b(\boldsymbol{\alpha}))^2 - A'(\boldsymbol{\alpha}) = \mathrm{Res}_{x_1}(A,B)(\boldsymbol{\alpha})/b(\boldsymbol{\alpha})^2 = 0.$$

Hence, we also have $\prod_{i=1}^{4} Q_i|_{x_1 = B'(\boldsymbol{\alpha})/b(\boldsymbol{\alpha})} = 0$. In other words that

$$\left(\frac{1}{b^4}\prod_{i=1}^{4}(B' \cdot b_i - b \cdot Q_i')\right)(\boldsymbol{\alpha}) = 0.$$

□

It follows that

$$\prod_{i=1}^{4}(B' \cdot b_i - b \cdot Q_i') \in \sqrt{\langle \mathrm{Res}_{x_1}(A,B)\rangle}.$$

In other words, for some positive integer k we have that $\mathrm{Res}_{x_1}(A,B)$ divides $\left(\prod_{i=1}^{4}(B' \cdot b_i - b \cdot Q_i')\right)^k$. As every irreducible factor of $\left(\prod_{i=1}^{4}(B' \cdot b_i - b \cdot Q_i')\right)^k$ is of degree3 or less, we get that, if the resultant is irreducible, then one of the multiplicands must be identically zero. Assume without loss of generality that $B'b_1 - bB_1' = 0$. It is not hard to verify that in this case either Q_1 is a scalar multiple of B and then Theorem3.1(prime-case) holds, or that B' is divisible by b. However, in the latter case it also holds that b divides the resultant, contradicting the assumption that it is irreducible. From now on we assume that $\mathrm{Res}_{x_1}(A,B)$ is reducible. We consider two possibilities. Either $\mathrm{Res}_{x_1}(A,B)$ has a linear factor or it can be written as

$$\mathrm{Res}_{x_1}(Q_1,Q_2) = C \cdot D,$$

for irreducible quadratic polynomials C and D. Consider the case where the resultant has a linear factor. If that linear factor is b, then b also divides B and Theorem3.1(product-case) holds. Otherwise, if it is a different linear form ℓ, then when setting $\ell = 0$ we get that the resultant of $A|_{\ell=0}$ and $B|_{\ell=0}$ is zero, and hence, either $B|_{\ell=0}$ is identically zero and Theorem3.1(product-case) holds or they share a common factor (see Theorem2.11). It is not hard to see that, if that common factor is of degree2, then Theorem3.1(product-case) holds, and if it is a linear factor, then Theorem3.1(linear-case) holds. Thus, the only

case left to handle (when $b \not\equiv 0$) is when there are two irreducible quadratic polynomials, C and D, such that $CD = \mathrm{Res}_{x_1}(A, B)$. As C and D divide two multiplicands in $\prod_{i=1}^{4}(B' \cdot b_i - b \cdot Q_i')$, we can assume, without loss of generality, that $(B' \cdot b_3 - b \cdot Q_3') \cdot (B' \cdot b_4 - b \cdot Q_4') \in \sqrt{\langle \mathrm{Res}_{x_1}(A, B) \rangle}$. Next, we express A', B', C and D as quadratics over b. That is,

$$\begin{aligned} A' &= \alpha b^2 + a_1 b + A'' \\ B' &= \beta b^2 + a_2 b + B'' \\ C &= \gamma b^2 + a_3 b + C'' \\ D &= \delta b^2 + a_4 b + D'', \end{aligned} \tag{3.2}$$

where $a_1, \ldots, D''$ do not involve b (nor x_1). We have the following two representations of the resultant:

$$\begin{aligned} \mathrm{Res}_{x_1}(A, B) &= B'^2 - b^2 \cdot A' \\ &= \beta^2 b^4 + 2\beta a_2 b^3 + (2\beta B'' + a_2^2) b^2 + 2a_2 B'' b + B''^2 - \alpha b^4 - a_1 b^3 - A'' b^2 \\ &= (\beta^2 - \alpha) b^4 + (2\beta a_2 - a_1) b^3 + (2\beta B'' + a_2^2 - A'') b^2 + 2a_2 B'' b + B''^2 \end{aligned} \tag{3.3}$$

and

$$\begin{aligned} \mathrm{Res}_{x_1}(Q_1, Q_2) &= CD \\ &= (\gamma b^2 + a_3 b + C'')(\delta b^2 + a_4 b + D'') \\ &= \gamma\delta b^4 + (\gamma a_4 + \delta a_3) b^3 + (\gamma D'' + a_3 a_4 + \delta C'') b^2 + (a_3 D'' + a_4 C'') b + C'' D''. \end{aligned} \tag{3.4}$$

Comparing the different coefficients of b in the two representations in equations(3.3) and(3.4), we obtain the following equalities:

$$B''^2 = C'' D'' \tag{3.5}$$

$$2a_2 B'' = a_3 D'' + a_4 C''. \tag{3.6}$$

We now consider the two possible cases giving equation(3.5).

1. Case1 explaining equation(3.5): After rescaling C and D, we have that $B'' = C'' = D''$. Equation(3.2) implies that for some linear form u, v we have that

   $$C = bv + B' \quad \text{and} \quad D = bu + B'.$$

   We now expand the resultant again:

   $$\begin{aligned} B'^2 + b(v+u)B' + b^2 vu &= (bv + B') \cdot (bu + B') = CD \\ &= \mathrm{Res}_{x_1}(A, B) = B'^2 - b^2 A'. \end{aligned}$$

   Hence,

   $$(v+u)B' + bvu = -bA'. \tag{3.7}$$

   Thus, either b divides B' in which case we get that b divides B and we are done as Theorem3.1(product-case) holds, or b divides $u + v$. That is,

   $$u + v = \varepsilon b \tag{3.8}$$

for some constant $\varepsilon \in \mathbb{C}$. Plugging this back into equation(3.7), we get

$$\varepsilon b B' + bvu = -bA'.$$

In other words, $\varepsilon B' + vu = -A'$. Consider the linear combination $Q = A + \varepsilon B$. We get that

$$\begin{aligned} Q = A + \varepsilon B &= (x_1^2 - A') + \varepsilon(x_1 b - B') \\ &= x_1^2 + \varepsilon x_1 b + vu \\ &= x_1^2 + x_1(u+v) + uv \\ &= (x_1 + u)(x_1 + v), \end{aligned} \tag{3.9}$$

where the equality in the third line follows from equation(3.8). Thus, equation(3.9) shows that some linear combination of A and B is reducible which implies that Theorem3.1(product-case) holds.

2. Case2 explaining equation(3.5): $B'' = u \cdot v$ and we have that, without loss of generality, $C'' = u^2$ and $D'' = v^2$ (where u, v are linear forms). Consider equation(3.6). We have that v divides $2a_2 B'' - a_3 D''$. It follows that v is also a factor of $a_4 C''$. Thus, either u is a multiple of v and we are back in the case where C'' and D'' are multiples of each other, or a_4 is a multiple of v. In this case, we get from equation(3.2) that for some constant δ',

$$D = \delta b^2 + a_4 b + D'' = \delta b^2 + \delta' v b + v^2.$$

Thus, D is a homogeneous polynomial in two linear forms. Hence, D is reducible, in contradiction.

This concludes the proof of Theorem3.1 for the case $b \not\equiv 0$.

Case $b \equiv 0$ To ease notation, let use denote $x = x_1$. We have that $A = x^2 - A'$ and that x does not appear in A', B. Let y be some variable such that $B = y^2 - B'$, and B' does not involve y (we can always assume this is the case without loss of generality). As before, we can subtract a multiple of B from A so that the term y^2 does not appear in A. If A still involves y, then we are back in the previous case (treating y as the variable according to which we take the resultant). Thus, the only case left to study is when there are two variables x and y such that

$$A = x^2 - A' \quad \text{and} \quad B = y^2 - B',$$

where neither A' nor B' involve either x or y. To ease notation, denote the rest of the variables as $\mathbf{z}$. Thus, $A' = A'(\mathbf{z})$ and $B' = B'(\mathbf{z})$. It is immediate that for any assignment to $\mathbf{z}$ there is an assignment to x, y that yields a common zero of A, B. By subtracting linear combinations of A and B from the Q_is, we can assume that for every $i \in [4]$

$$Q_i = \alpha_i xy + a_i(\mathbf{z})x + b_i(\mathbf{z})y + Q_i'(\mathbf{z}).$$

We next show that, under the assumptions in the theorem's statement, it must be the case that either A' or B' is a perfect square or that $A' \sim B'$. In either situation, we have that Theorem3.1(product-case) holds. We first show that, if A' and B' are linearly independent, then this implies that at least one of A', B'

is a perfect square. Let $Z(A,B)$ be the set of common zeros of A and B, and denote by $\pi_{\mathbf{z}} : Z(A,B) \to \mathbb{C}^{n-2}$, the projection on the $\mathbf{z}$ coordinates. Note that $\pi_{\mathbf{z}}$ is surjective; as for any assignment to $\mathbf{z}$, there is an assignment to x, y that yields a common zero of A, B.

Claim (Author Claim3.4)**.** *Let $Z(A,B) = \bigcup_{i=1}^{k} X_k$ be the decomposition of $Z(A,B)$ to irreducible components. Then there exists $i \in [k]$ such that $\pi_{\mathbf{z}}(X_i)$ is dense in $\mathbb{C}^{n-2}$.*

Proof. $\bigcup_{i=1}^{k} \pi_{\mathbf{z}}(X_i) = \pi_{\mathbf{z}}(Z(A,B)) = \mathbb{C}^{n-2}$, as $\pi_{\mathbf{z}}$ is a surjection, it holds that $\bigcup_{i=1}^{k} \overline{\pi_{\mathbf{z}}(X_i)} = \mathbb{C}^{n-2}$. We also know that $\mathbb{C}^{n-2}$ is irreducible, and thus, there is $i \in [k]$ such that $\overline{\pi_{\mathbf{z}}(X_i)} = \mathbb{C}^{n-2}$, which implies that $\pi_{\mathbf{z}}(X_i)$ is dense. □

Assume, without loss of generality, that $\pi_{\mathbf{z}}(X_1)$ is dense. We know that $X_1 \subseteq Z(\prod_{i=1}^{4} Q_i)$, so we can assume, without loss of generality, that $X_1 \subseteq Z(Q_1)$. Observe that this implies that Q_1 must depend on at least one of x, y. Indeed, if Q_1 depends on neither, then it is a polynomial in $\mathbf{z}$, and hence, its set of zeros cannot be dense. Every point $\boldsymbol{\xi} \in X_1$ is of the form $\boldsymbol{\xi} = (\delta_1\sqrt{A'(\boldsymbol{\beta})}, \delta_2\sqrt{B'(\boldsymbol{\beta})}, \boldsymbol{\beta})$, for some $\boldsymbol{\beta} \in \mathbb{C}^{n-2}$, $\delta_1, \delta_2 \in \{\pm 1\}$ (δ_1, δ_2 may be functions of $\boldsymbol{\beta}$). Thus, $Q_1(\boldsymbol{\xi}) = Q_1(\delta_1\sqrt{A'(\boldsymbol{\beta})}, \delta_2\sqrt{B'(\boldsymbol{\beta})}, \boldsymbol{\beta}) = 0$, and we obtain that

$$\alpha_1\delta_1\delta_2\sqrt{A'(\boldsymbol{\beta}')}\sqrt{B'(\boldsymbol{\beta}')} + a_1(\boldsymbol{\beta}')\delta_1\sqrt{A'(\boldsymbol{\beta}')} + b_1(\boldsymbol{\beta}')\delta_2\sqrt{B'(\boldsymbol{\beta}')} + Q_1'(\boldsymbol{\beta}') = 0. \tag{3.10}$$

As we assumed that Q_1 depends on at least one of x, y, let us assume without loss of generality that either α_1 or a_1 are nonzero. The next argument is similar to the proof that $\sqrt{2}$ is irrational. Note that we use the fact that $\delta_1^2 = \delta_2^2 = 1$.

$$\begin{aligned}
(3.10) \implies & B'(\boldsymbol{\beta}')\left(\alpha_1\delta_1\sqrt{A'(\boldsymbol{\beta}')} + b_1(\boldsymbol{\beta}')\right)^2 = \left(Q_1'(\boldsymbol{\beta}') + a_1(\boldsymbol{\beta}')\delta_1\sqrt{A'(\boldsymbol{\beta}')}\right)^2 \\
\implies & B'(\boldsymbol{\beta}')\left(\alpha_1^2 A'(\boldsymbol{\beta}') + 2\delta_1\alpha_1 b_1(\boldsymbol{\beta}')\sqrt{A'(\boldsymbol{\beta}')} + b_1(\boldsymbol{\beta}')^2\right) \\
& \qquad = Q_1'(\boldsymbol{\beta}')^2 + 2\delta_1 a_1(\boldsymbol{\beta}')Q_1'(\boldsymbol{\beta}')\sqrt{A'(\boldsymbol{\beta}')} + a_1(\boldsymbol{\beta}')^2 A'(\boldsymbol{\beta}').
\end{aligned}$$

$$\begin{aligned}
\implies & \delta_1\sqrt{A'(\boldsymbol{\beta}')}\left(2\alpha_1 b_1(\boldsymbol{\beta}')B'(\boldsymbol{\beta}') - 2a_1(\boldsymbol{\beta}')Q_1'(\boldsymbol{\beta}')\right) \\
& \qquad = Q_1'(\boldsymbol{\beta}')^2 + a_1(\boldsymbol{\beta}')^2 A'(\boldsymbol{\beta}') - B'(\boldsymbol{\beta}')\left(\alpha_1^2 A'(\boldsymbol{\beta}') + b_1(\boldsymbol{\beta}')^2\right) \qquad (3.11) \\
\implies & A'(\boldsymbol{\beta}')\left(2\alpha_1 b_1(\boldsymbol{\beta}')B'(\boldsymbol{\beta}') - 2a_1(\boldsymbol{\beta}')Q_1'(\boldsymbol{\beta}')\right)^2 \\
& \qquad = \left(Q_1'(\boldsymbol{\beta}')^2 + a_1(\boldsymbol{\beta}')^2 A'(\boldsymbol{\beta}') - B'(\boldsymbol{\beta}')\left(\alpha_1^2 A'(\boldsymbol{\beta}') + b_1(\boldsymbol{\beta}')^2\right)\right)^2. \qquad (3.12)
\end{aligned}$$

This equality holds for every $\boldsymbol{\beta} \in \pi_{\mathbf{z}}(X_1)$, which is a dense set, and hence holds as a polynomial identity. Thus, either $A'(\mathbf{z})$ is a square, in which case we are done or it must be the case that the following identities hold

$$Q_1'(\mathbf{z})^2 + a_1(\mathbf{z})^2 A'(\mathbf{z}) - B'(\mathbf{z})\left(\alpha_1^2 A'(\mathbf{z}) + b_1(\mathbf{z})^2\right) = 0 \tag{3.13}$$

and

$$\alpha_1 b_1(\mathbf{z})B'(\mathbf{z}) - a_1(\mathbf{z})Q_1'(\mathbf{z}) = 0. \tag{3.14}$$

By symmetry, if $B'(\mathbf{z})$ is not a square (as otherwise we are done), we get that

$$\alpha_1 a_1(\mathbf{z})A'(\mathbf{z}) - b_1(\mathbf{z})Q_1'(\mathbf{z}) = 0. \tag{3.15}$$

If $\alpha_1 = 0$, then we get from(3.14) that $Q_1' \equiv 0$. Hence, by(3.13),

$$a_1(\mathbf{z})^2 A'(\mathbf{z}) = B'(\mathbf{z})b_1(\mathbf{z})^2.$$

Since we assumed that A' and B' are independent, this implies that A' and B' are both squares. If $Q_1' \not\equiv 0$ (and in particular, $\alpha_1 \neq 0$), then either $a_1(\mathbf{z}) = b_1(\mathbf{z}) \equiv 0$, in which case Equation(3.13) implies that $Q_1'(\mathbf{z})^2 = \alpha_1^2 A'(\mathbf{z})B'(\mathbf{z})$, and we are done (as either both A' and B' are squares or they are both multiples of Q_1'), or equations(3.14) and(3.15) imply that $\alpha_1^2 A'(\mathbf{z})B'(\mathbf{z}) = Q_1'(\mathbf{z})^2$ which again implies the claim. This concludes the proof of Theorem3.1 for the case $b \equiv 0$ and thus the proof of the theorem. □

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